Agent-Computer Observation Interfaces Enable Dynamic Computer Use

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Abstract

SWE-agent [Yang et al., 2024] showed that the agent-computer *action* interface is an underexplored design axis for software-engineering agents. We show the same for the *observation* interface in computer-use (CU) agents: getting it right unlocks dynamic computer use from existing CU models without retraining—though, as our later model studies show, the optimal mix of observation components is model-specific. Current CU agents—closed and open-source alike—observe the world through periodic screenshots every 3–5 s and have no audio perception, leaving them blind to video playback, animations, transient UI events, meetings, notifications, and spoken instructions. We introduce the **Agent-Computer Observation Interface (AOI)**, a model-agnostic perception layer that decouples observation (continuous, adaptive) from action (discrete). Its three gated components—inter-step keyframe capture, volume-gated audio transcription, and CU-model-generated visual narration that persists as text—produce almost nothing on static, silent content, reducing the AOI to the standard loop.

On **DynaCU-Bench**, our benchmark of 100 dynamic browser tasks across 10 categories (plus a 50-task static control), eight CU models—closed and open-source, from 7B to frontier scale—gain +17 to +48 pp over their screenshot-only baselines with zero retraining (paired McNemar $p < 10^{-3}$ for every model); the strongest, Claude Sonnet 4.6, reaches a 3-seed mean of 78.7% (best seed 82%). On a 12-task audio-focused subset, two production streaming baselines—OpenAI Realtime [OpenAI, 2024] (3/12) and Gemini Live [Google, 2025] (0/12)—fall far short of the AOI-equipped agent (12/12).

Three findings reframe CU perception. *Selection doesn't matter, and keyframes pay off only through narration*: uniform, random, pixel-difference, and CLIP keyframe selection are statistically indistinguishable, so no learned or semantic frame selector is needed; and the keyframe images add just +1 pp as raw visual input but +10 pp once the model narrates them into persistent text—a keyframe×narration synergy concentrated on the transient-UI and carousel tasks where content changes between steps. *Narration helps mainly by making the model articulate what it sees*: visual narration is the largest of the three gated perception components (+18 pp), mostly an inference-time articulation effect, with persistent text memory load-bearing on a minority of multi-step tasks. *The observation interface is not a fixed bundle*: on Gemini 3 Flash the keyframe stream—never regressive on the earlier model generations we measured—flips to **−12** pp, a regression a causal probe traces to image-token dilution rather than content; audio and the structured scaffold stay positive, and dropping keyframes recovers +21 pp over standard. Observation-interface components must therefore be selected per model: the observation interface is a separable design axis along which models converge unevenly per component.

1 Introduction

The agent-computer observation interface as a design axis. SWE-agent [Yang et al., 2024]'s core contribution was not a new model but a careful design of the agent-computer *action* interface—

which commands the model is given, how inputs and outputs are formatted, what feedback they return. Small changes there dramatically affect what a fixed model can accomplish, making the action interface a separable, underexplored design axis. We make the analogous argument for the *observation* interface in computer-use (CU) agents. Every CU agent we are aware of—closed-source [Anthropic, 2024a, OpenAI, 2025, Google DeepMind, 2025, Andreux et al., 2025] and open-source [Wang et al., 2025a,b, Awadallah et al., 2025, Song et al., 2025]—ties observation to action: one screenshot per action step. We argue that decoupling observation (continuous, adaptive) from action (discrete) is the single change that unlocks dynamic computer use from existing CU models with zero retraining—with the caveat, developed in Section 8, that which components help is model-specific.

The current state. On OSWorld [Xie et al., 2024], screenshot-based agents have moved from sub-10% at introduction to over 50% in 2025 (Surfer 2 [Andreux et al., 2025]: 50.7%; UI-TARS-2 [Wang et al., 2025a]: 47.5%), closing much of the gap to the 72.4% human baseline. These agents can fill forms, navigate websites, edit documents, and use professional software.

The blind spot. All these agents observe through *discrete static screenshots* every 3–5 seconds, and are entirely deaf. Between screenshots, they are blind. They cannot understand on-screen video, catch transient UI events that auto-dismiss between observations, hear meeting speech or notification sounds, or perceive animations. A comprehensive survey [Sager et al., 2025] catalogues many open challenges but does not identify the observation modality as a gap. Quantitative evidence is mounting: GUI-World [Chen et al., 2024] shows video LLMs struggle across temporal GUI tasks; VideoWebArena [Jang et al., 2024] reports that long-context models perform *worse* with video than without; VideoGameBench [Zhang et al., 2025] finds even paused-game evaluation yields only 1.6% completion; CUA-Suite [Chen et al., 2025] catalogues 55 hours of temporal dynamics that screenshot-based agents miss entirely.

Why existing approaches fall short. Retraining CU models on video data is expensive, model-specific, and undemonstrated at scale. Video-native models (Gemini) amount to brute-force frame sampling that wastes tokens on redundant static frames. Native real-time multimodal models (Gemini Live [Google, 2025], OpenAI Realtime [OpenAI, 2024]) work but lock users into a single provider, offer no selective perception, and require API migration; we evaluate them in Section 9. Cradle [Tan et al., 2024]'s pixel-level frame differencing is the closest predecessor but lacks audio perception, visual narration, and a structured observation record.

Our approach. The **Agent-Computer Observation Interface (AOI)** is a lightweight perception layer that converts continuous screen and audio streams into the sparse images and text that existing CU models already accept. It sits between the environment and any image-based CU model; on static, silent content its gates produce nothing and behaviour reverts to the standard loop; on dynamic content it produces extra keyframes, audio transcripts, and CU-model-generated narration that persists as text after the source images are pruned.

Contributions.

1. We identify the **observation interface** as a critical but overlooked CU design axis, complementary to SWE-agent's action interface, and articulate *decoupling observation from action* as the operative design principle.
2. We introduce the **Agent-Computer Observation Interface (AOI)**, a model-agnostic perception layer with three gated components (inter-step keyframe capture, volume-gated audio scene understanding, CU-model-generated visual narration).
3. We release **DynaCU-Bench** (100 dynamic tasks) and a **Static-50** no-degradation baseline.
4. Eight CU models equipped with the AOI and **zero retraining** gain +17 to +48 pp on dynamic tasks (paired McNemar $p < 10^{-3}$ for every model): six in the main table plus a two-model Qwen3-VL replication (Table 4); three additional later-released models (Gemini 3 Flash, Grok-

4.3, Grok-4-fast-reasoning) are reported as case studies in Section 8.

5. A controlled *narration-discarded* ablation plus a content audit decompose the visual-narration component's +18 pp gain: most of it is inference-time articulation (a chain-of-thought effect), with persistent text memory load-bearing on the minority of tasks whose answer spans multiple steps (Section 6.2).

2 Background: The CU Agent Loop

All current CU agents follow the same observe–reason–act loop:

$$\texttt{repeat:}\quad s_t \leftarrow \mathrm{Screenshot}(),\quad a_t \leftarrow \mathrm{CU\_Model}(s_t, \tau),\quad \mathrm{Execute}(a_t),\quad \mathrm{Wait}(\delta) \tag{1}$$

where τ is the trajectory history and δ is a post-action buffer (typically 500 ms). The interval between observations is determined by model inference time + action execution + buffer, typically 3–5 seconds total. Everything that occurs between screenshots—visual changes, audio events, transient UI elements—is invisible to the agent.

Some models support multi-image input through screenshot history (UI-TARS: up to 5 images, Claude CU: ∼10), but all images are post-action screenshots from prior steps. None captures what happens *between* steps.

Four categories of content are systematically missed: (i) *transient UI events*—toast notifications and dialogs that appear and vanish within one observation interval; (ii) *continuous visual media*—video playback, animated tutorials, transitioning slides; (iii) *audio events*—meeting speech, notification sounds, error alerts; (iv) *periodic visual noise*—loading spinners and blinking cursors that should be *ignored* but trigger false positives in naive frame differencing.

3 System Design

The AOI sits between the environment and any existing CU model, observing the entire interval between agent steps and providing the model with additional keyframes, audio descriptions, and accumulated text context—all in the standard image + text format that every CU model already accepts. For static, silent tasks, the AOI produces nothing and behaviour is identical to the standard loop; for dynamic tasks, the model receives strictly more information. No retraining is needed.

3.1 Architecture Overview

The AOI has three components, each preceded by a fast gate (sub-millisecond) that short-circuits processing when there is nothing to perceive; Figure 1 contrasts the resulting pipeline with the standard CU loop. For a frame that passes the gates, the additional cost is dominated by CLIP-ViT-B/16 (∼7 ms on GPU) for visual frames and Whisper large-v3 (variable, on demand) for audio segments; for the static, silent frames that dominate most workloads the per-frame cost is the sub-millisecond gate alone.

(1) **Inter-step keyframe capture**: continuous screen sampling at ∼3 Hz with gating to suppress redundant frames. Produces 0–5 keyframe images per step.

(2) **Volume-gated audio observer**: RMS energy gate followed by ASR (Whisper large-v3). Produces a text transcription per step, only when audio is present.

(3) **Visual narration context**: the CU model itself generates a brief text description of new visual information as a side-output of each inference call. These narrations accumulate in the trajectory and persist after keyframe images are pruned from context.

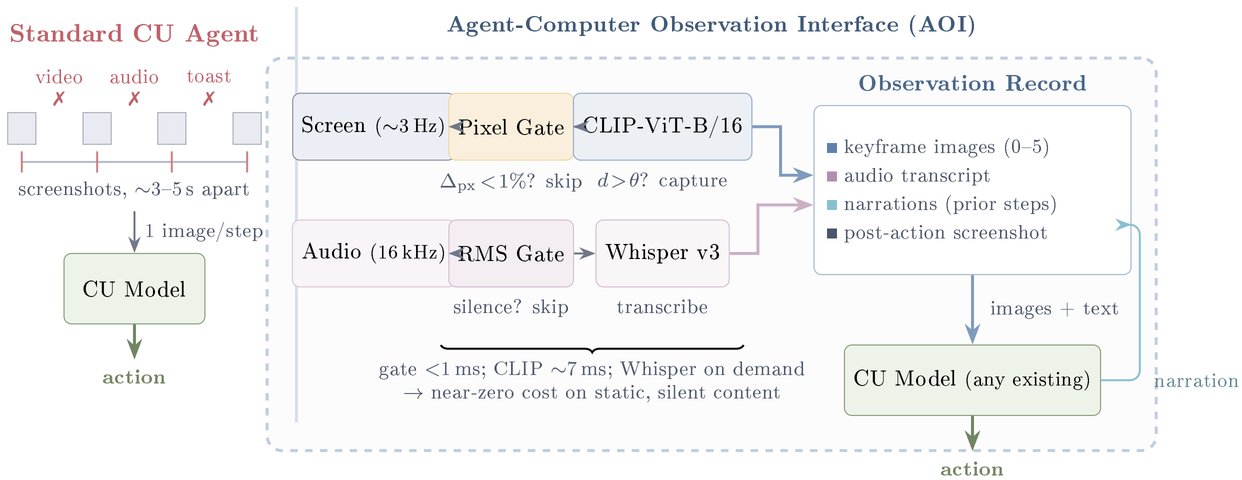


Figure 1: **Standard CU agents vs. AOI-equipped agents.** *Left:* Current agents observe through periodic screenshots (∼3–5 s apart) and are blind and deaf between observations—missing video, audio, and transient UI events (✗). *Right:* The AOI continuously monitors screen and audio streams through fast gates (<1 ms) that skip processing for static, silent content. When gates fire, keyframe extraction (shown here with optional CLIP filtering; see Section 6.1) and Whisper ASR produce images and text for the observation record. Visual narrations feed back as persistent text memory. The AOI wraps any existing CU model with zero retraining.

3.2 Inter-Step Keyframe Capture

Between agent steps, the screen may change (a video plays, a dialog appears, a slide transitions) or stay static. The AOI continuously samples the screen at ∼3 Hz and selects a subset of frames to include as keyframe images in the observation record (up to 5 per step). Multiple selection strategies are possible—uniform fixed-rate sampling, pixel-level differencing, random sampling, or semantic filtering via CLIP embeddings [Radford et al., 2021]. We evaluate five variants spanning these four families (uniform at 1 and 3 FPS, pixel-diff, random, CLIP) in Section 6.1 and find that all converge to similar accuracy, so the default implementation uses simple pixel-change gating: if fewer than 1% of pixels changed since the last captured frame, the sample is skipped (cost: <1 ms on CPU). For static screens, no keyframes are produced and the component has zero overhead. Algorithm 1 formalises the two-stage CLIP variant; the default selector drops the CLIP stage (lines 5–10) and uses only Stage 1.

3.3 Volume-Gated Audio Observation

Current CU agents have zero audio perception. They cannot hear meeting speech, notification sounds, or error alerts. Pure ASR models (Whisper) transcribe speech but produce nothing for non-speech audio events.

Volume gate. In most computer use—file editing, form filling, silent web browsing—there is no audio. Computing RMS energy of the audio buffer is sub-millisecond. If below the silence threshold, the audio model call is skipped entirely. Cost: ∼0 ms for the majority of steps.

When audio is present, we use Whisper large-v3 [Radford et al., 2023] for speech transcription, operating on 16 kHz mono audio captured from the system audio output via PulseAudio virtual devices. The audio buffer spans the full inter-step interval including the post-action buffer, capturing action feedback sounds (click confirmations, error beeps).

Algorithm 1 Two-stage adaptive keyframe extraction (CLIP variant). This algorithm is shown for completeness; the selection-method ablation in Section 6.1 finds it statistically indistinguishable from the simpler pixel-only gate (Stage 1) and uniform / random sampling, so the default selector in every main result is Stage 1 alone.

Require: Screen sample x_t, anchor embedding e_{anchor}, pixel threshold α, CLIP threshold θ

$\Delta_{\text{px}} \leftarrow \text{PixelChangeRatio}(x_t, x_{\text{prev}})$
if $\Delta_{\text{px}} < \alpha$ **then**
  return Skip {Stage 1: no pixel change}
end if
$e_t \leftarrow \text{CLIP}(x_t)$
$d \leftarrow 1 - \cos(e_t, e_{\text{anchor}})$
if $d > \theta$ **then**
  $e_{\text{anchor}} \leftarrow e_t$ {Re-anchor}
  return CaptureKeyframe(x_t, t)
end if
return Skip {Stage 2: below semantic threshold}

Overlapping windows. Agent step boundaries are determined by model inference latency, not by speech content. A typical 3.5-second audio chunk captures ∼8–9 words at normal speaking rate, almost always falling mid-sentence. To resolve boundary artifacts, the audio buffer includes ∼3.5 seconds of overlap from the previous interval, providing acoustic continuity across step boundaries.

3.4 Visual Narration for Long-Term Context

CU models retain only the most recent images in context. Keyframes from earlier steps are pruned. For tasks spanning many steps, the agent loses all visual memory of earlier content. Audio transcriptions persist naturally as text in the trajectory, but visual observations have no analogous persistence.

At each step, the CU model outputs both an action and a brief visual narration—a text description of what is visually new in the current keyframes. This narration is generated in the same inference call that produces the action—adding no extra inference call, only a small number of output tokens—and persists indefinitely in the trajectory, even after the corresponding images are pruned. Following the caption-then-reason principle of LLoVi [Zhang et al., 2023], narrations are generated by the CU model itself (no separate captioner) and are task-relevant rather than generic.

3.5 Observation Record

At each step, the CU model receives a structured document combining text context from recent prior steps (audio transcriptions, visual narrations, actions taken) and raw observations from the current interval (new audio transcription, any captured keyframe images, and the post-action screenshot). For static, silent tasks, this document contains only the post-action screenshot—identical to the standard loop. Figure 2 shows a concrete example of the document format; Figure 3 grounds the pipeline in real pixels and verbatim narrations from a recorded benchmark run.

3.6 Implementation

The AOI is implemented in Python (∼2,600 lines) as a modular layer that wraps any CU model (full software/hardware/hyperparameter details in Appendix D). Screen capture uses a headless browser

Observation Record — Step N (Meeting Task B-E1)

```
# CONTEXT -- text from prior steps (no images)
Step N-1 (t ≈ 18.5-22.0 s):
AUDIO: "Here's the team.  As you can see we
have strong cross-functional coverage."
VISUAL: "Meeting slide shows Project Team
with 6 members listed in a grid."
ACTION: wait()

# NEW -- current interval observations
Step N (t ≈ 22.0-25.5 s):
AUDIO: "And I want to mention, we've moved
the launch date to April 28th.
Please update your calendars."
[IMAGE: post-action screenshot]

# TASK
Read the meeting slides and listen to the
discussion.  What is the new launch date?
Enter it in the Launch Date field.
```

Figure 2: Example observation record sent to the CU model at step N of a meeting task. The **context** section provides text from prior steps (audio transcriptions and visual narrations persist after images are pruned). The **new** section contains the current audio transcription and the post-action screenshot (image). In this case, no keyframes were captured (the slide did not change), so only the screenshot is included as an image. The task instruction is appended at the end.

(Playwright) at ∼3 Hz. Audio capture and injection use PulseAudio virtual devices (virtual speaker and monitor source) at 16 kHz mono. CLIP-ViT-B/16 runs on GPU (∼150 MB VRAM). Whisper large-v3 runs as a separate HTTP service on CPU to avoid GPU contention with the CU model. Audio synthesis for benchmark tasks uses the Web Speech API (`speechSynthesis`).

CU models are accessed through their native APIs. Cloud models (Claude Sonnet 4.6, GPT-5.4, Gemini 2.5 Flash) use HTTP APIs. Local models (EvoCUA-32B, Fara-7B) are served via vLLM with 4-bit quantization on a single NVIDIA RTX PRO 6000 (96 GB VRAM).

4 DynaCU-Bench

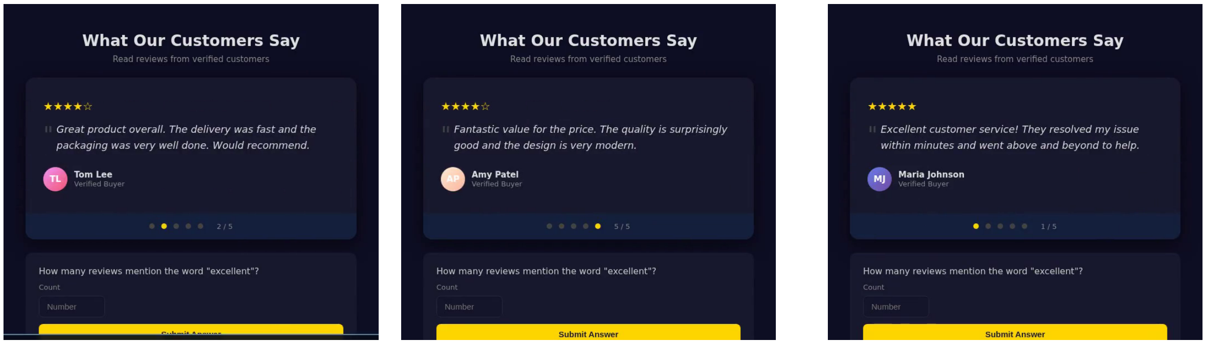
4.1 Design Principles

Existing CU benchmarks [Xie et al., 2024, Zhou et al., 2023, He et al., 2024] consist almost entirely of tasks solvable from static screenshots. We introduce **DynaCU-Bench**, a benchmark of 100 browser-based tasks that specifically require dynamic visual and/or audio perception—tasks that are unsolvable from static screenshots alone.

Design principles: (i) tasks must *require* temporal visual or audio perception; (ii) tasks represent realistic computer-use scenarios, not artificial constructs; (iii) tasks execute in a reproducible headless browser environment; (iv) tasks cover all four dynamic content categories from Section 2; (v) tasks include difficulty stratification (3 easy, 4 medium, 3 hard per category).

4.2 Task Categories

DynaCU-Bench spans 10 categories across three capability axes: audio perception (A), visual-temporal perception (V), and real-time interaction (I). Each category has 10 tasks; representative



step 1: testimonial 2/5 (Tom Lee) step 2: testimonial 5/5 (Amy Patel) step 3: testimonial 1/5 (Maria Johnson)

Visual narrations accumulated in the trajectory (verbatim from the evaluation log)

`Step 1 (0 keyframes): "The carousel shows testimonial 2/5 from Tom Lee with a 4-star rating about fast delivery and good packaging, with no mention of 'excellent'." ACTION: click(567, 400)`

`Step 2 (1 keyframe): "The carousel now shows testimonial 5/5 from Amy Patel with a 4-star rating about fantastic value, quality, and modern design -- no mention of 'excellent'." ACTION: click(567, 401)`

`Step 3 (2 keyframes): "The carousel has cycled back to testimonial 1/5 from Maria Johnson with 'Excellent customer service!' -- and the transition keyframe also shows a Sarah Robinson testimonial partially visible..." ACTION: fill(#answerInput, "2")` ✓

Figure 3: **What the agent actually sees: real frames and narrations from a recorded benchmark run** (task D-E2, "How many reviews mention the word 'excellent'?", Claude Sonnet 4.6 + AOI full—the run reported in Table 3; 3 steps, passed). The testimonial carousel rotates every few seconds, so no single screenshot can show more than one of the five reviews. *Top:* screen states observed across the three agent steps (frames from the run recording, recorder status bar cropped). *Bottom:* the visual narrations the CU model generated at each step, quoted verbatim from the evaluation log. The narrations persist as text in the trajectory, letting the agent accumulate evidence across carousel states and answer "2" at step 3.

examples per category appear in Appendix B.

4.3 Evaluation Protocol

Each task executes in a headless Chromium browser (Playwright) with PulseAudio virtual devices for audio I/O. Audio content is delivered via the Web Speech API (`speechSynthesis`) with edge-TTS for high-quality voices. Agent audio input is injected into a virtual microphone via `pacat`.

Three evaluation modes are used: *DOM-based* (93 tasks): a deterministic JavaScript function `getTaskResult()` returns the expected value; *LLM-judged* (1 task): a judge model scores the agent's response against a rubric; *Hybrid* (6 tasks): DOM gate (did the agent act?) combined with LLM quality assessment. Tasks have a maximum of 15 agent steps and a per-task wall-clock budget of `duration_s + 10 s + AOI overhead`, where `duration_s` is the task's stimulus duration (typically 15–90 s, scaling with difficulty); whichever limit is reached first ends the run. These wall-clock budgets matter only for the slowest models (e.g. Grok-4 at ∼80–180 s/call; Section 8.1).

Table 1: DynaCU-Bench task categories. A = audio perception, V = visual-temporal perception, I = real-time interaction. Each category contains 3 easy, 4 medium, and 3 hard tasks.

Cat.	Domain	Axes	Description
A	Podcast comprehension	A	Listen to audio narration, extract facts or answer questions
B	Meeting participation	A+V+I	Follow meeting with slides and speech, take notes or act
C	Video/screencast	V	Watch terminal recordings or IDE demos with auto-advancing frames
D	Carousel/animation	V	Perceive CSS transitions, rotating content, product carousels
E	Live dashboard	V	Monitor real-time updating dashboards with `setInterval` changes
F	Transient UI events	V	Respond to toast notifications, auto-dismissing banners
G	Phone/voice call	A+I	Listen to `speechSynthesis` calls and respond
H	Interview/voice	A+I	Answer questions via `SpeechRecognition`, complete interviews
I	Collaborative editing	V+I	Handle remote changes in collaborative document editors
J	Browser games	V+I	Play interactive games requiring temporal attention

5 Evaluation

5.1 Setup

CU models. We evaluate six models spanning the capability and scale spectrum: (i) Claude Sonnet 4.6 [Anthropic, 2024a] (closed-source); (ii) GPT-5.4 [OpenAI, 2026] (closed-source; the latest GPT-5 series tool-calling-capable vision model exposed by OpenAI's chat-completions API at evaluation time); (iii) Gemini 2.5 Flash [Google DeepMind, 2025] (closed-source); (iv) Grok-4 [xAI, 2026] (closed-source, xAI; reported in Section 8.1 with attention to the inference-latency confound); (v) EvoCUA-32B [Meituan, 2026] (open-source, 32B; weights and inference recipe released through Meituan's GitHub); (vi) Fara-7B [Awadallah et al., 2025] (open-source, 7B). Two further open-source models—Qwen3-VL-235B-A22B-Instruct and Qwen3-VL-30B-A3B-Instruct [Qwen Team, 2025], accessed through OpenRouter—are evaluated on the same standard vs. AOI full contrast as a replication (Table 4). We use each model's vendor-recommended decoding settings (temperature, top-p) unchanged. We do not rely on any vendor-reported OSWorld score for any claim in this paper; we report only the numbers measured under our own DynaCU-Bench harness.

Observation configurations. The primary contrast is **Standard** (one screenshot per step, no audio—the current paradigm) vs. **AOI (full)** (inter-step keyframes + Whisper ASR + visual narration), evaluated with all six models. The remaining modes are ablations and diagnostic controls, evaluated with Claude Sonnet 4.6 unless stated otherwise and introduced in detail where first used; because they recur across several sections, Table 2 collects every observation mode used in the paper in one place.

Table 2: Glossary of all observation modes used in the paper. KF = inter-step keyframes (with selection strategy); Audio = Whisper transcripts of system audio; Narr. = CU-model-generated visual narration persisted as text; [PAGE EL] = DOM element list in the prompt; Traj. = structured prior-step trajectory wrapping. **Standard** and **AOI full** are run on all models; other modes are Claude Sonnet 4.6 ablations, with two exceptions: the selection ablation is also replicated on Qwen3-VL-32B, and `standard_audio`, `aoi_audio`, and the keyframe probes are Gemini 3 Flash diagnostics.

Mode	**KF**	**Audio**	**Narr.**	**[PAGE EL]**	**Traj.**	**Introduced**
Standard	—	—	—	—	✓	Table 3
standard_minimal	—	—	—	—	—	§7.3
standard_pageel_only	—	—	—	✓	—	§7.3
standard_structured	—	—	—	✓	✓	§7.2
standard_audio	—	✓	—	—	✓	§8.2
Selection variants (uniform/pixel/random)	variant	—	—	✓	✓	§6.1
AOI (visual only)	CLIP	—	—	✓	✓	§6.1
AOI (visual + ASR)	CLIP	✓	—	✓	✓	§6.1
aoi_audio	—	✓	✓	✓	✓	§8.2
AOI (full)	pixel gate	✓	✓	✓	✓	Table 3
AOI full, narr. discarded	pixel gate	✓	not persisted	✓	✓	§6.2
Keyframe probes (max$k$/noise/dup/reorder)	modified	✓	✓	✓	✓	§8.3

5.2 Main Results

Table 3 and Figure 4 present the main results. The AOI produces large, consistent gains across all six models, all highly significant by paired McNemar's exact test ($p < 10^{-3}$ for every model; $p < 10^{-9}$ for Claude and Gemini, $p < 10^{-8}$ for EvoCUA). *Note on statistical power:* Each configuration is evaluated once on 100 binary-outcome tasks (no repeated trials). We use paired McNemar's exact mid-p tests on the discordant tasks to assess significance and report 95% Wilson confidence intervals on the marginal success rates. Pairwise differences between selection methods (e.g., 56% vs. 58%) are not significant ($p > 0.5$ for every pair, see Table 9); the large between-tier gaps are.

Claude Sonnet 4.6 improves from 38% to 82% (+44 pp) with the full AOI. This is the strongest absolute result. The ablation modes reveal that all inter-step visual capture methods converge to 56–58% (a +18–20 pp gain over the raw-screenshot baseline, regardless of selection method—though, as Sections 6.1 and 6.3 show, this transition also turns on the structured prompt scaffold: most of the jump is the scaffold, with the keyframe images contributing +10 pp in context through narration), adding ASR reaches 64%, and visual narration provides the largest single-component content gain, reaching 82% (mechanism decomposed in Section 6.2).

GPT-5.4 improves from 37% to 57% (+20 pp). The gain is consistent with Claude's but the absolute ceiling is lower, suggesting GPT-5.4's CU capabilities are the binding constraint once perception is adequate.

Gemini 2.5 Flash scores 21% on the standard baseline despite being a video-native model with built-in streaming capabilities. This confirms that video-native architectures do not automatically solve the observation problem—the standard CU agent loop still constrains Gemini to periodic screenshots. With the AOI, Gemini improves to 69% (+48 pp)—the largest absolute gain of any model tested. Section 7 (Table 16) decomposes this gain into +25 pp from the AOI's structured prompt format and +23 pp from new perception: a video-native model benefits from *both* better presentation of existing context and explicit between-step perception.

EvoCUA-32B improves from 18% to 55% (3× improvement, +37 pp). This open-source 32B model, which barely functions on dynamic tasks without the AOI, becomes competitive with GPT-

Table 3: DynaCU-Bench results: task success rate (%) on the 100 dynamic tasks. Brackets show 95% Wilson confidence intervals. **Bold** indicates the best result per model. Δ is the absolute gain over the standard baseline; p-values are paired McNemar's exact mid-p tests. All ablation modes are tested on Claude Sonnet 4.6. "Claude 4.6" abbreviates Claude Sonnet 4.6 throughout. Grok-4 is the slowest model at inference (∼80–180 s/call), which holds its absolute scores down even when AOI unblocks perception; see Section 8.1.

Configuration	**Claude 4.6**	**GPT-5.4**	**Gemini 2.5**	**Grok-4**	**EvoCUA-32B**	**Fara-7B**
Standard (screenshot)	38 [29.1, 47.8]	37 [28.2, 46.8]	21 [14.2, 30.0]	4 [1.6, 9.8]	18 [11.7, 26.7]	17 [10.9, 25.5]
Uniform 1 FPS	58 [48.2, 67.2]	—	—	—	—	—
Uniform 3 FPS	56 [46.2, 65.3]	—	—	—	—	—
Pixel-diff only	58 [48.2, 67.2]	—	—	—	—	—
Random keyframes	56 [46.2, 65.3]	—	—	—	—	—
AOI (visual only)	58 [48.2, 67.2]	—	—	—	—	—
AOI (visual + ASR)	64 [54.2, 72.7]	—	—	—	—	—
AOI (full)	**82** [73.3, 88.3]	**57** [47.2, 66.3]	**69** [59.4, 77.2]	**25** [17.5, 34.4]	**55** [45.2, 64.4]	**34** [25.5, 43.7]
Δ (full − std)	+44	+20	+48	+21	+37	+17
p (McNemar)	1.3×10^{-10}	8.8×10^{-5}	2.9×10^{-12}	3.0×10^{-6}	3.0×10^{-9}	4.9×10^{-4}

Table 4: Additional open-source replication on the full 100-task DynaCU-Bench: Qwen3-VL family via OpenRouter, evaluated with the same harness as the six models in Table 3. p-values are paired McNemar exact tests on the discordant tasks (b/c = success only under standard / only under AOI). Both gains fall inside the +17 to +48 pp band of the main table.

Model	**Standard**	**AOI full**	**Δ (pp)**	b/c	p **(McNemar)**
Qwen3-VL-235B-A22B-Instruct	22	**64**	+42	4/46	4.5×10^{-10}
Qwen3-VL-30B-A3B-Instruct	18	**42**	+24	8/32	1.8×10^{-4}

5.4's AOI-equipped performance.

Fara-7B improves from 17% to 34% (+17 pp), doubling its score. As the smallest model (7B parameters), it is most constrained by reasoning capacity, but still benefits substantially from improved perception.

Figure 5 makes the mechanism behind these gains concrete with two side-by-side trajectories: on a meeting task the standard agent waits out its entire step budget because it cannot hear the spoken answer, while the AOI agent transcribes it and acts in two steps; on a video task the AOI agent's narration captures the answer in a single step.

Additional open-source replication (Qwen3-VL family). To probe the model-agnostic claim further, we run the same standard vs. AOI full contrast on two more open-source vision-language models via OpenRouter (provider: DeepInfra): **Qwen3-VL-235B-A22B-Instruct** [Qwen Team, 2025], the largest open-source VLM MoE available at evaluation time, and the smaller **Qwen3-VL-30B-A3B-Instruct**. Both land inside the +17 to +48 pp band of Table 3 (Table 4): +42 pp on the 235B model (22 → 64, McNemar $p = 4.5 \times 10^{-10}$) and +24 pp on the 30B model (18 → 42, $p = 1.8\times10^{-4}$). The 235B model's AOI full score (64) exceeds GPT-5.4's (57), and within the Qwen family the AOI gain grows with scale (+24 vs. +42 pp), consistent with the reasoning-capacity bottleneck observed on Fara-7B. (Two further candidates, UI-TARS-1.5-7B and GLM-4.5V, were unavailable through our OpenRouter provider list at evaluation time.)

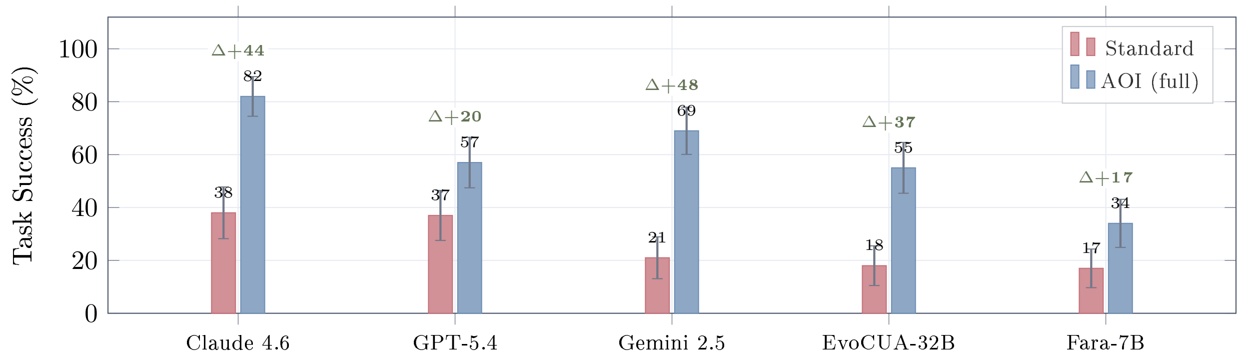


Figure 4: **Main results on DynaCU-Bench (100 tasks).** The AOI produces large gains across CU models without any retraining. Green numbers show absolute improvement in percentage points. Gemini 2.5 Flash shows the largest absolute gain (+48 pp) and relative gain (3.3×), while Claude achieves the highest absolute score (82%, seed 1; 3-seed mean 78.7%). Five of the six models tested are plotted here; Grok-4 (standard 4, AOI full 25, Δ+21 pp) is omitted from the chart for legibility and reported separately in Table 19 (Section 8.1). Error bars are approximate 95% intervals using a symmetric Gaussian span; exact (asymmetric) Wilson intervals are tabulated in Table 3 and used for all reported p-values.

Table 5: Three-seed replication of the headline Claude Sonnet 4.6 AOI full configuration on the 100 dynamic tasks. Seed 1 is the run reported in Table 3; seeds 2–3 are fresh re-runs at the API's default sampling parameters (Anthropic's Messages API does not expose a deterministic seed, so each call samples fresh). The 82% headline is at the upper edge of a 76–82% range, with 3-run mean 78.7% and standard deviation 3.1 pp. The McNemar tests in Table 3 compare within-seed paired outcomes and remain valid for the standard-vs.-AOI contrast; the seed std reported here is the additional uncertainty on the AOI full absolute score.

Configuration	**Seed 1**	**Seed 2**	**Seed 3**	**Mean**	**Std**
Claude Sonnet 4.6 + AOI full	82	78	76	78.7	3.1

Variance: three-seed replication of the headline. Each Table 3 cell is one trial of 100 binary tasks at the model's default sampling temperature, folding decoding variance into "task hardness" without separating the two. To quantify the decoding component, we re-run Claude Sonnet 4.6 + AOI full for two additional seeds using the same harness; results are in Table 5.

The decoding-variance question is independent of the McNemar statistical-power question: McNemar quantifies whether two configurations differ *given a fixed run*; the seed sweep quantifies how much a single run's score moves under fresh decoding noise.

5.3 Per-Category Analysis

Table 6 and Figure 6 reveal category-level patterns:

Audio categories (A, B, G) show the largest gains. Standard baselines score 0–2/10 on podcast, meeting, and phone tasks across the five tabulated models—these tasks are impossible without audio perception. The AOI unlocks 3–10/10 depending on the model, with Claude achieving near-perfect scores (9/10 or 10/10).

Visual-temporal categories (C, D, F) benefit consistently. Video, carousel, and transient UI tasks improve by 1–9 points across models, with a single small regression on Fara-7B's F (4 →

Task B-E1: Meeting — "What is the new launch date?"

Standard (screenshot-only) — FAIL

15 steps, 51.2 s

1. `wait()` — sees static slide
2. `wait()` — same slide
3. `wait()` — same slide
4. `wait()` — same slide
5. `wait()` — same slide

⋮

15. `wait()` × step budget exhausted

Agent never hears the spoken date.
Waits all 15 steps without ever acting.

AOI (full) — PASS

2 steps, 26.2 s

1. `wait()`
   Audio: "Here's the team. As you can see we have strong cross-functional coverage."
   Narration: "Meeting slide shows Project Team with 6 members."
2. `fill(#launchDate, "April 28th")` ✓
   Audio: "We've moved the launch date to April 28th. Please update."
   Narration: "Audio announced the new launch date."

Agent hears "April 28th" and fills the form immediately.

Task C-E1: Video — "What was the first command in the terminal recording?"

Standard — PASS (6 steps)

1. `wait()` — recording playing
2. `wait()` — missed first command
3. `wait()` — scrolling through
4. `wait()`
5. `wait()`
6. `fill(...)` ✓ lucky timing

AOI — PASS (1 step)

1. `fill(#answerInput, "git clone ...")` ✓
   Narration: "The terminal recording shows the first command was git clone. . . "

Narration captures the command from the recording immediately.

Figure 5: Trajectory comparison on two representative tasks (*illustrative; cherry-picked to expose the mechanism*). **Top:** Meeting task (B-E1) where the launch date is spoken aloud. The standard agent cannot hear it and exhausts its step budget on 15 futile `wait()` steps. The AOI agent transcribes the audio, hears "April 28th," and acts in 2 steps. **Bottom:** Video task (C-E1) where the answer appears in a terminal recording. The standard agent needs 6 steps with lucky timing; the AOI agent's narration captures the content in 1 step.

3). Category E (dashboard) shows smaller gains because dashboard tasks often have sufficient information in single screenshots—the dynamic updates are periodic and the current value is always visible.

Category J (games) is an exception. Claude's score drops from 5 to 2 with the AOI. Browser games are bound by action latency, not perception: category J has the highest step count (118 steps, 11.8 per task) and high keyframe density (0.55 KF/step; Table 14), so CLIP encoding and keyframe processing add ∼2–3 s per step on top of the model's ∼3 s inference latency—roughly doubling the agent's effective reaction time and causing it to miss action windows it would otherwise meet. The AOI is designed for human-paced computer use, not real-time interactive applications.

EvoCUA-32B shows disproportionate gains on audio tasks. The model jumps from 0/10 to 8/10 on both podcast (A) and phone (G) categories, suggesting that audio perception particularly benefits models whose visual grounding is weaker.

Table 6: Per-category task success (pass/10) for standard vs. AOI full across the five models with full per-category breakdowns; Grok-4 is excluded here for legibility because its absolute scores are latency-limited (see Section 8.1, Table 19). Categories are grouped by primary perception axis: A = audio, V = visual-temporal, I = real-time interaction.

		Claude 4.6		**GPT-5.4**		**Gemini 2.5**		**EvoCUA-32B**		**Fara-7B**	
Cat.	**Axes**	Std	AOI	Std	AOI	Std	AOI	Std	AOI	Std	AOI
A (Podcast)	A	0	**9**	0	3	0	**8**	0	**8**	0	3
B (Meeting)	A+V+I	2	**10**	2	3	2	**8**	1	6	1	5
C (Video)	V	4	**9**	5	**8**	5	6	3	6	3	5
D (Carousel)	V	4	**9**	6	7	4	7	3	4	1	3
E (Dashboard)	V	8	8	6	**9**	1	**9**	2	4	2	2
F (Transient)	V	4	**9**	2	**8**	0	**9**	0	5	4	3
G (Phone)	A+I	1	**9**	1	4	0	**8**	0	**8**	1	4
H (Interview)	A+I	7	**9**	6	7	5	7	6	**8**	1	3
I (Collab)	V+I	3	**8**	4	5	2	6	2	5	2	4
J (Games)	V+I	5	2	5	3	2	1	1	1	2	2
Total		38	**82**	37	**57**	21	**69**	18	**55**	17	**34**

Table 7: Task success by difficulty level (standard → AOI full). Easy: 30 tasks (3 per category). Medium: 40 tasks (4 per category). Hard: 30 tasks (3 per category).

	Standard			**AOI Full**		
Model	Easy	Med	Hard	Easy	Med	Hard
Claude 4.6	17/30	15/40	6/30	**28**/30	**34**/40	**20**/30
GPT-5.4	15/30	17/40	5/30	21/30	26/40	10/30
Gemini 2.5	10/30	9/40	2/30	24/30	33/40	12/30
EvoCUA-32B	6/30	8/40	4/30	20/30	26/40	9/30
Fara-7B	9/30	5/40	3/30	19/30	11/40	4/30

5.4 Difficulty Breakdown

Table 7 shows that the AOI lifts all difficulty levels. For Claude, easy task success rises from 57% to 93%, medium from 38% to 85%, and hard from 20% to 67%. The pattern is consistent across models: the AOI unlocks perception-gated tasks at every difficulty, but hard tasks remain challenging because they require both perception *and* sophisticated reasoning. Fara-7B's small gain on medium/hard (5→11, 3→4) compared to its large gain on easy (9→19) suggests that reasoning capacity, not perception, becomes the bottleneck for the 7B model on harder tasks.

5.5 Efficiency

Table 8 reveals two distinct patterns—one for token cost, one for wall-clock latency. *Token cost falls for every cloud model*: Claude \$2.72 → \$1.35 (−50%), GPT-5.4 \$3.23 → \$2.76 (−15%), Gemini 2.5 Flash \$0.32 → \$0.16 (−50%). This is because better perception leads to fewer steps per task: Claude uses 4.8 steps with AOI vs. 10.7 without, Gemini drops from 11.5 to 5.4, and EvoCUA-32B from 9.3 to 5.3. The observation overhead (12–32 s per task) is more than offset by saving 2–6 model calls per task.

Wall-clock latency, however, does not always fall. For four of five models, total task time

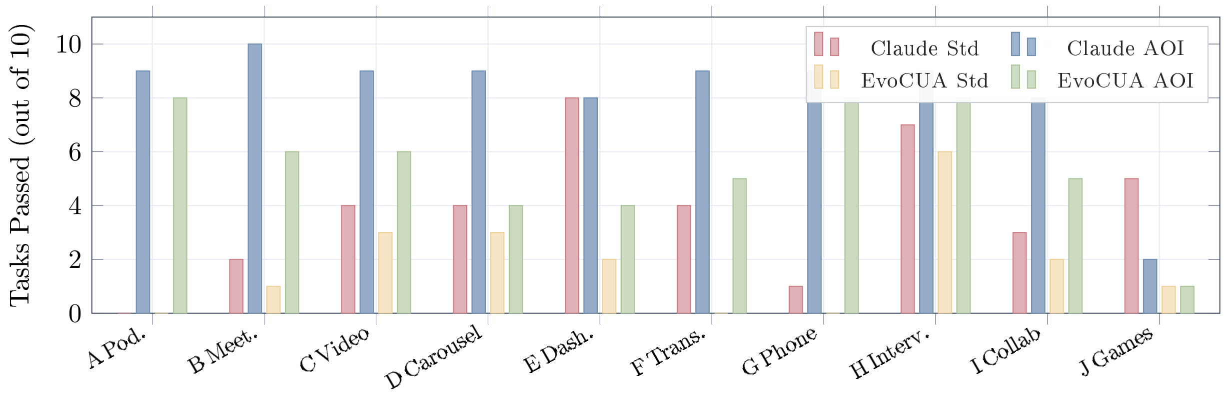


Figure 6: **Per-category results for Claude Sonnet 4.6 and EvoCUA-32B.** Audio-dependent categories (A, B, G) show the largest gains—from near-zero to 6–10/10. Visual-temporal categories (C, D, F) improve consistently. Category J (games) is the exception: real-time action latency, not perception, is the bottleneck. Category E (dashboard) shows minimal change for Claude because current values are visible in single screenshots.

rises with the AOI: Claude 40.6 s → 46.2 s (+14%), GPT-5.4 26.6 s → 50.6 s (+90%), Gemini 55.3 s → 61.9 s (+12%), and Fara-7B 35.1 s → 90.7 s (+158%). Only EvoCUA-32B sees absolute time *decrease* (117.0 s → 113.9 s, −3%) because its very high per-step model latency (∼10 s/step) means step reduction dominates the added observation overhead. Summary: the AOI is consistently cheaper in dollars, but for fast-inference models it is slower in wall-clock time because per-step observation work exceeds the latency saved per skipped step. For batch automation or cost-sensitive production the AOI is unambiguously preferred; for latency-sensitive interactive use, the trade-off is per-model.

6 Component Analysis: Ablations and Mechanisms

6.1 Ablation: Observation Mode Analysis

Table 9 and Figure 7 reveal the contribution of each AOI component (the complete per-category breakdown for all eight ablation modes is in Appendix A, Table 26):

Selection method does not affect accuracy (56–58%). The five visual-only keyframe strategies (Section 3.2: uniform 1 and 3 FPS, pixel-diff, random, and the two-stage CLIP filter of Algorithm 1) converge to a narrow band of 56–58%, with no significant pairwise McNemar difference (every $p > 0.5$; Table 11). Practitioners can therefore use whatever keyframe strategy fits their latency and compute budget; for the CLIP variant, a sweep of the distance threshold θ across a 15× range shows no significant sensitivity either (Appendix C, Figure 9).

The keyframe images pay off only through narration. The +18–20 pp lift from the raw `standard` baseline to any keyframe tier ($p = 8.8 \times 10^{-5}$ for Standard → Uniform 1 FPS) overstates the keyframe images' direct value: this transition also turns on the AOI's structured prompt scaffold (the `[PAGE ELEMENTS]` DOM list and prior-step trajectory wrapping), which `standard` lacks and which Section 7.2 shows carries +19 pp on its own for Claude (38 → 57). Holding the scaffold fixed, the keyframe *images* on their own add only +1 pp (`standard_structured` 57 → AOI visual-only 58, with no audio or narration). But that is not the deployed configuration. When the model also narrates what it sees, the same keyframe images are worth +10 pp: an `aoi_audio` run (structured scaffold + ASR + narration, but *no* keyframes) scores 72/100, and adding the keyframe stream back

Table 8: Efficiency and cost across observation configurations on the 100-task benchmark for the five models with full instrumented runs. Grok-4 is excluded because its ∼80–180 s/call latency floor distorts the cost/latency picture (Section 8.1). Steps = mean steps per task (lower is better); Time = mean wall-clock per task; Lat. = total model inference latency per task; Obs = total observation processing time per task, including screenshot capture (∼2 s/step) plus CLIP encoding, keyframe selection, and audio processing for AOI configurations. Tokens are accounted as input + output per task, estimated from the evaluation logs by `experiments/compute_tokens.py`: text at ∼1.3 tokens per word and a flat 258 tokens per image (Anthropic's vision-token approximation for downscaled screenshots; actual vision tokenization varies by provider). The same estimator is applied to every configuration, so relative comparisons are meaningful while absolute counts are approximate. $ / 100 tasks computed from these estimates at May 2026 list prices: Claude Sonnet 4.6 $3/Mtok in, $15/Mtok out; GPT-5.4 $5/$15; Gemini 2.5 Flash $0.30/$2.50; EvoCUA-32B and Fara-7B run locally. Despite per-step observation overhead, the AOI reduces *cost* by 15–50% across the three cloud models (50% on Claude, 15% on GPT-5.4, 50% on Gemini 2.5 Flash) because the step count drops more than the per-step input grows.

Configuration	Steps	Time (s)	Lat. (s)	Obs (s)	Tokens (k/task)	$/100 tasks
Claude Sonnet 4.6						
Standard	10.7	40.6	26.3	2.5	7.6	$2.72
AOI full	**4.8**	46.2	15.2	12.0	**3.8**	**$1.35**
GPT-5.4						
Standard	10.0	26.6	14.4	2.4	5.8	$3.23
AOI full	**7.6**	50.6	11.2	17.8	**5.0**	**$2.76**
Gemini 2.5 Flash						
Standard	11.5	55.3	40.4	2.6	7.6	$0.32
AOI full	**5.4**	61.9	20.6	20.6	**4.0**	**$0.16**
EvoCUA-32B (local)						
Standard	9.3	117.0	99.0	2.3	5.8	—
AOI full	**5.3**	113.9	72.0	22.6	**4.0**	—
Fara-7B (local)						
Standard	13.4	35.1	24.0	2.5	9.2	—
AOI full	**9.3**	90.7	17.0	31.6	**7.1**	—

recovers the full 82 (Section 6.3, Table 13). The keyframe channel is thus *synergistic with narration*—worth +1 pp as raw images but +10 pp once the model converts the captured frames into persistent text—and that +10 pp concentrates on exactly the categories where content changes *between* steps (transient-UI F +4, carousel D +2), not on the audio or static-dashboard categories where one screenshot per step already suffices. The selection-invariance finding is unaffected: whatever the keyframe channel contributes, it does so regardless of how frames are chosen.

Open-source replication on Qwen3-VL-32B. To rule out a Claude-specific phenomenon, we replicate the convergence on Qwen3-VL-32B-Instruct [Qwen Team, 2025] via OpenRouter (substituting for EvoCUA-32B and Fara-7B, which could not be re-run on our host; Appendix D), using the three cheapest selection methods (Uniform 1 FPS, Pixel-diff, Random) on the 50 visual-temporal tasks (categories C, D, E, F, J)—the subset where keyframe selection actually differs. The three methods again cluster within a narrow band with no pairwise McNemar significance ($p > 0.4$ for every pair; Table 10): the convergence is mechanism-driven—any inter-step frame breaks the agent's blindness—not a quirk of a single model.

Table 9: Ablation study on Claude Sonnet 4.6: contribution of each AOI component. All configurations include per-category breakdown for categories most affected. Small per-category drops between adjacent tiers (e.g. B: 5→4 from visual-only to visual+ASR; C: 10→9 from visual+ASR to AOI full) are within sampling noise (no McNemar significance); per-tier transitions on the 100-task total are tested in Table 11.

Mode	**Total** /100	**A**	**B**	**C**	**G**	**H**
		(pass/10, selected categories)				
Standard	38	0	2	4	1	7
Uniform 1 FPS	58	0	4	**10**	1	8
Uniform 3 FPS	56	2	5	8	1	7
Pixel-diff	58	1	5	9	1	8
Random keyframes	56	1	6	8	1	8
AOI visual only	58	1	5	9	1	8
AOI visual + ASR	64	1	4	**10**	4	**9**
AOI full	**82**	**9**	**10**	9	**9**	**9**

Table 10: Selection-method ablation on an open-source vision model (Qwen3-VL-32B-Instruct via OpenRouter), 50 visual-temporal tasks (categories C, D, E, F, J). The three selection methods cluster within a narrow band, mirroring the Claude Sonnet 4.6 pattern. Pairwise McNemar tests find no significant differences ($p > 0.4$ for every pair), confirming that the convergence finding is not Claude-specific.

Selection method	**Pass / 50**	**Vs. adjacent**
Uniform 1 FPS	26	—
Pixel-diff	27	vs. Uniform: p =0.69
Random keyframes	27	vs. Pixel-diff: p =1.00

Adding ASR (+6 pp). Whisper speech transcription lifts performance from 58% to 64%. Taken alone this transition does not reach significance at N=100 ($p = 0.18$, Table 11), so we report it as directional; the audio channel's contribution is confirmed independently at scale in the four-way Gemini 3 decomposition (Section 8.2) and by the audio-category gains in Table 6. The gain concentrates on audio-dependent categories (G: 1→4, H: 8→9). Categories A and B show minimal improvement because podcast and meeting comprehension require understanding the *content* of speech over multiple steps, not just detecting that speech occurred.

Full AOI (+18 pp over visual+ASR; $p = 5.3 \times 10^{-4}$). The full AOI with visual narration reaches 82%. The large gap from 64% to 82% makes visual narration the largest single component; Section 6.2 decomposes the mechanism—mostly inference-time articulation of what is on screen, with persistent text memory load-bearing on tasks whose answer spans multiple steps. The improvement is especially pronounced on podcast (A: 1→9) and meeting (B: 4→10) tasks, where narration forces the agent to articulate and accumulate information across many steps.

Resolving the CoT-vs-memory confound. The +18 pp gain could in principle conflate two effects: the *persistent text memory* that survives keyframe pruning, and a *chain-of-thought* effect from generating the narration itself. Section 6.2, next, disentangles the two with a *narration-discarded* ablation.

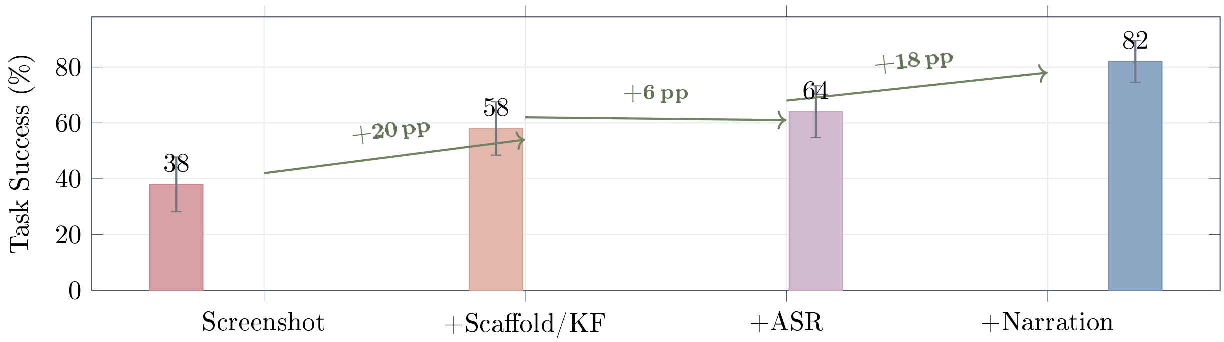


Figure 7: **Ablation: progressive contribution of each AOI component** (Claude Sonnet 4.6). The first step (+20 pp; any selection method achieves it, Table 9) *bundles* inter-step keyframes with the structured prompt scaffold that bare `standard` lacks; the scaffold carries +19 of it, and the keyframe images add +1 pp as raw input but +10 pp once narration is present (Sections 6.1, 6.3). ASR adds +6 pp (directional; $p = 0.18$ for this transition alone, Table 11). Visual narration adds +18 pp—the largest single-component content gain; Section 6.2 decomposes its mechanism. Error bars are approximate 95% intervals using a symmetric Gaussian span (exact Wilson intervals in Table 3).

Table 11: Pairwise McNemar's exact mid-p tests between successive ablation tiers on the same 100 tasks (Claude Sonnet 4.6). Each row tests whether the labelled configuration differs significantly from the previous one. "b" / "c" are the discordant counts (success in only the previous / only the current configuration). The selection-method group is statistically indistinguishable; the between-tier transitions are highly significant.

Comparison	**Total**	b	c	p
Standard → Uniform 1 FPS	38 / 58	3	23	8.8×10^{-5}
Uniform 1 FPS → Uniform 3 FPS	58 / 56	7	5	0.77
Uniform 3 FPS → Pixel-diff	56 / 58	4	6	0.75
Pixel-diff → Random keyframes	58 / 56	4	2	0.69
Random keyframes → CLIP adaptive	56 / 58	4	6	0.75
CLIP adaptive → AOI visual+ASR	58 / 64	4	10	0.18
AOI visual+ASR → AOI full	64 / 82	4	22	5.3×10^{-4}

6.2 Narration: Persistent Memory or Chain-of-Thought?

The full AOI's largest single gain comes from visual narration (+18 pp). This component does two things at once: it (i) asks the CU model to articulate the new visual content as a side-output during inference, and (ii) persists that text in the trajectory so it survives keyframe pruning. To separate these effects we run the *narration-discarded* ablation: the model is prompted exactly as in AOI full and produces a narration each step (so any inference-time chain-of-thought benefit is preserved), but the narration string is dropped from the trajectory before it can be reused as context. The keyframe and audio pipelines are otherwise identical to AOI full.

Table 12 reports the result. The narration-discarded score is 74/100, sitting between the visual+ASR baseline (64) and the full AOI (82). The +18 pp gap therefore decomposes into: (i) a +8 pp contribution from persisting the narration as text (74 → 82, McNemar $p = 0.039$), which is the *only significant component at* N=100; and (ii) a directional +10 pp from inference-time chain-of-thought during narration generation (64 → 74, $p = 0.12$), which is consistent in sign and magnitude

Table 12: Narration ablation on Claude Sonnet 4.6 (100 dynamic tasks). McNemar's exact mid-p tests on the discordant tasks. The narration-discarded configuration sits between the two endpoints, indicating that both the inference-time chain-of-thought effect (+10 pp from ASR to ND) and the narration-persistence effect (+8 pp from ND to full) contribute to the full +18 pp gain. Only the persistence effect reaches significance at N=100; a content audit of the discordant tasks (body text) finds literal memory recall in 2 of the 10 wins, the rest operating through within-step inputs and trajectory-level priming.

Configuration	**Total**	A	B	C	D	E	F	G	H	I	J	**Vs. adjacent**
AOI visual+ASR (no narration)	64	1	4	10	9	9	9	4	9	6	3	—
AOI full *narration discarded*	74	9	10	8	8	7	4	9	9	7	3	vs. ASR: $p = 0.12$
AOI full (narration as persistent memory)	**82**	9	10	9	9	8	**9**	9	9	8	2	vs. disc.: $p = 0.039$

but underpowered—reaching $p < 0.05$ at this effect size would require $N \sim 300$.

Narration content audit: how much of the +8 pp is actually memory? The McNemar test establishes that persisting the narration helps; it does not show *why*. To audit the mechanism we examine the 10 tasks where `aoi_full` passes and narration-discarded fails. For each, we extract the narrations from *prior* steps—the persistent text memory available to the agent at the moment of the winning action—and ask an external Claude Opus 4.7 judge to classify whether those prior narrations explicitly contain the load-bearing fact the winning action depends on (audit script and judgments are released with the code). Only **2 of the 10** (20%) audited wins show the prior narrations explicitly containing the answer (a *memory hit*): C-H3 (video breakpoints, where prior narrations recorded breakpoint locations from earlier video frames) and C-M2 (sequential video page titles, recorded as the pages advanced). In the remaining 8 cases the load-bearing information is available *within the winning step itself* (current screenshot, current audio, or the narration generated at that same step); three of the eight have zero prior narrations because the winning action lands at the first step.

The refined reading: because the narration-discarded run changes *only* whether the narration text persists, the +8 pp (74 → 82, $p = 0.039$) is causally attributable to persistence—the audit refines the persistence effect's *mechanism*, not its existence. Explicit memory recall (prior narrations literally containing the load-bearing fact) is the mechanism on 2 of the 10 discordant wins, both on genuinely multi-step tasks (C-H3 video breakpoints, C-M2 sequential page titles). On the other 8 the load-bearing fact is present within the winning step itself, so persistence cannot be supplying a verbatim answer there; it instead helps by steering earlier steps onto a better trajectory, and on the three first-step wins (no prior narration exists yet) the discordance is pure decoding variance. The full +18 pp visual-narration contribution is therefore best summarised as $\sim +10$ pp from inference-time chain-of-thought (directional, $p = 0.12$) plus a real +8 pp persistence effect ($p = 0.039$) whose mechanism is literal memory recall on the minority of multi-step tasks and weaker trajectory-level priming elsewhere. Narration's primary mechanism is making the model *articulate* what it sees; its persistence matters most where information genuinely spans steps.

6.3 Keyframes Pay Off Through Narration: the `aoi_audio` Decomposition

Section 6.1 showed that, holding the structured scaffold fixed, the keyframe *images* on their own add only +1 pp on Claude (`standard_structured` 57 → `aoi_visual` 58). That measurement, however, switches off audio and narration—not the configuration in which the AOI is deployed. To measure the keyframe images' marginal value *in context*, we run `aoi_audio` (structured scaffold + ASR +

Table 13: Keyframe images' marginal value *in context*: `aoi_audio` (scaffold + ASR + narration, *no* keyframes) vs. `aoi_full` (keyframes added back), 100 dynamic tasks. Δ_{KF} is the keyframe-image contribution with audio and narration both present. Contrast with the +1 pp the same images add without narration (Section 6.1): the keyframe channel is synergistic with narration on Claude and the most model-dependent AOI component overall, spanning +10 to −12 pp across four models (cf. the −12 pp on Gemini 3, traced to image-token dilution in Section 8.3). †GPT-5.4 is measured through OpenRouter (both modes, same adapter) because its direct OpenAI key lacked the inference scope at revision time; its `aoi_full` runs higher here than the OpenAI-direct 57 of Table 3 (a backend/snapshot difference), so only its within-backend Δ_{KF} is comparable to the other rows.

Model	`aoi_audio` **(no KF)**	`aoi_full` **(+KF)**	$\Delta_{\mathbf{KF}}$ **(pp)**
Claude Sonnet 4.6	72	**82**	+10
Gemini 2.5 Flash	63	**69**	+6
GPT-5.4†	**72**	70	−2
Gemini 3 Flash	**57**	45	−12

visual narration, with the keyframe stream disabled) and compare it to `aoi_full` (which adds the keyframes back); the difference isolates the keyframe images with audio and narration both present. Crucially, `aoi_audio` still has the model narrate—it narrates the single post-action screenshot each step—so the contrast asks specifically whether the *extra inter-step frames* carry information beyond one screenshot per step.

Table 13 reports the result. On Claude the keyframe images are worth +10 pp in context (72 → 82)—an order of magnitude more than the +1 pp they add as raw input without narration. The two measurements together identify a **keyframe×narration synergy**: the captured inter-step frames matter not as extra pixels the policy attends to directly, but as raw material the model converts into persistent text. Narration of one screenshot per step (`aoi_audio`) already recovers most of the perception gain; the extra inter-step frames add a further +10 pp, and that increment concentrates on exactly the categories where content changes *between* steps—transient-UI F (+4) and carousel D (+2)—and is near-zero on the audio (A, B, G, H) and static-dashboard (E) categories where one screenshot per step already suffices. Gemini 2.5 Flash shows the same pattern at smaller magnitude (63 → 69, +6 pp, again concentrated on F +3 and D +2). GPT-5.4 sits at the neutral point (72 → 70, −2 pp, within sampling noise): the inter-step frames neither help nor hurt once it is already narrating. On Gemini 3 Flash the same contrast runs the other way (`aoi_audio` 57 vs. `aoi_full` 45, −12 pp): the identical image stream that helps Claude through narration overwhelms Gemini 3's action grounding (Section 8.3). The keyframe-image channel is thus the most model-dependent AOI component, spanning a +10 pp gain on Claude and +6 pp on Gemini 2.5 (both realised through narration), through GPT-5.4's neutral −2 pp, to a −12 pp *penalty* on Gemini 3—while audio transcription and narration, both delivered as *text*, are robustly positive across every model. This sharpens the operative rule of Section 8.4: deliver observation to the model as text (transcripts and narration) wherever possible, and treat the raw keyframe-image stream as a per-model toggle.

6.4 Per-Step Observation Statistics

Table 14 and Figure 8 show the per-step observation statistics for the AOI. The data confirms the gate-suppression design: audio-only categories (G, H) produce zero keyframes, while visual-only

Table 14: Per-step observation statistics for **Claude Sonnet 4.6 + AOI full**, broken down by task category. % KF = percentage of steps where at least one keyframe was captured. KF/Step = average keyframes per step. % Audio = percentage of steps with audio transcription. Audio-heavy categories (A, B, G, H) show 26–74% audio activation. Visually dynamic categories (D, J) average 0.55–1.38 keyframes per step. Audio-only categories (G, H) produce zero keyframes—the pixel and CLIP gates correctly suppress extraction when only audio changes.

Cat.	Domain	Steps	% KF	Tot. KF	KF/Step	% Audio
A	Podcast	35	8.6	3	0.09	42.9
B	Meeting	34	50.0	18	0.53	73.5
C	Video	39	28.2	12	0.31	2.6
D	Carousel	61	60.7	84	1.38	0.0
E	Dashboard	52	28.8	18	0.35	0.0
F	Transient	38	15.8	7	0.18	0.0
G	Phone	38	0.0	0	0.00	26.3
H	Interview	27	0.0	0	0.00	37.0
I	Collab	39	10.3	4	0.10	17.9
J	Games	118	36.4	65	0.55	0.0
All		**481**	**28.3**	**211**	**0.44**	**14.1**

categories (D, E, F, J) produce zero audio activations. Category D (carousel) has the highest keyframe density at 1.38 per step because carousel animations produce frequent visual changes. Category A (podcast) has the lowest keyframe rate (0.09) because podcast tasks involve a mostly static UI with audio content. Overall, only 28.3% of steps produce keyframes and 14.1% activate audio processing—the gates are effective at suppressing unnecessary computation.

6.5 Cross-Engine ASR Validation (DynaCU-Real-Local)

To verify that the AOI's audio pipeline does not silently fail under a different acoustic profile than the edge-TTS audio used in DynaCU-Bench, we re-ran 12 audio tasks with audio rendered by `espeak` (a formant synthesizer) and screencasts driven by real `asciinema` v2 cast files. Standard and AOI full both score 11/12 (the same single failure for both, and pretraining suffices for the rest; Table 27 in Appendix E), so the result is *not* a perception comparison—it is a no-degradation check: Whisper produces intelligible transcripts from the espeak audio in every case (verified by manual inspection of the transcripts). Designing real-recording tasks whose answers are not recoverable from CU-model pretraining is harder than it sounds and is left to follow-up work (Section 11). Harness details, per-task breakdown, and asset sources are in Appendix E.

7 Decomposing the Gain: Prompt Format vs. Perception

7.1 Static-50: No Degradation on Static Work

A key design claim is that the AOI does not degrade performance on purely static, silent desktop work. We verify this on **Static-50**, a 50-task baseline (22 easy, 16 medium, 12 hard) of pure HTML tasks: form filling, table reading, calculation, sequence puzzles, and policy reasoning. None of the pages contain animations, timers, video, or audio; the rendered DOM is identical at $t = 0$ and $t = \infty$. This is large enough to detect a 5–10 pp difference between modes with reasonable power; Table 15 reports the result.

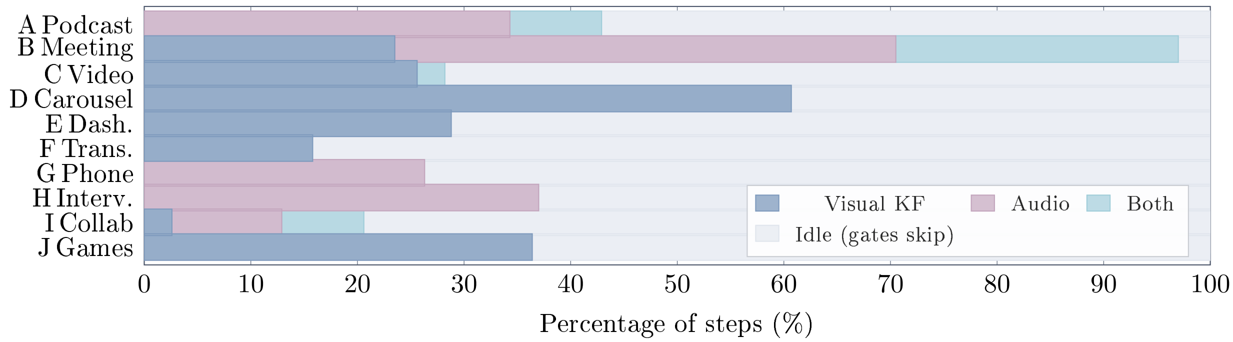


Figure 8: **Observation activity per category.** Stacked bars show the percentage of steps in which each observation channel fires. The gray "idle" portion is steps where both gates suppress processing. Categories cluster: D is highly visual; G and H are audio-only; B uses both. Weighted by step count, ∼61% of all steps are idle.

Table 15: Static-50 verification: Claude Sonnet 4.6 on 50 pure-HTML tasks (no dynamic content). The AOI's gates suppress almost all extraction—7 keyframes across all 50 tasks (all incidental UI feedback such as form-field focus changes) and 0 audio segments: the perception pipeline runs but produces almost no observations, validating the gates' suppression behaviour. The AOI does not degrade static work; its higher score is a prompt-format effect (see body text).

Mode	**Pass/50**	**Avg Steps**	**Total KF**	**Audio Seg.**
Standard	43 (86.0%)	3.26	0	0
AOI full	**50** (100.0%)	**1.62**	7	0

Across all 50 AOI-mode runs, the pixel gate captured only 7 keyframes total (0.09 per step, all incidental UI feedback such as form-field focus or checkbox state changes) and the volume gate never fired. The AOI scores 50/50 vs. Standard's 43/50. The 7 Standard failures are all multi-element form tasks (S-E3 dropdown selection, S-E4 contact form, S-E9 reference-code copy, S-M2 address form, S-M8 settings toggles, S-M9 meeting schedule, S-H5 email reply)—exactly the workload where the structured observation record's DOM element list and prior-step text trajectory help most. A perfect score on a workload with near-zero perception extraction is consistent with a prompt-format effect, not a hidden perception gain.

7.2 Prompt Format vs. Perception: A Direct Measurement

The +14 pp Static-50 advantage is therefore a *prompt-format* effect—the AOI's structured observation record (DOM element list, prior-step trajectory summary, explicit context/new sectioning) over a raw screenshot—not a perception gain. We measure this directly on DynaCU-Bench via the `standard_structured` mode, which supplies the AOI's prompt scaffolding while disabling all keyframe and audio extraction: its gain over Standard is the prompt-format effect, and AOI full minus standard_structured is the perception-only effect.

Table 16 reports the decomposition for five models (three cloud + two open-source); Gemini 3 Flash is deferred to Table 20 (Section 8.2). Both Δ_{prompt} and Δ_{percept} are positive for every model: the structured prompt format and the perception channel each contribute independently rather than one re-explaining the other. The split varies with model—Claude Sonnet 4.6 is perception-leaning (+25 vs. +19 pp), Gemini 2.5 Flash and GPT-5.4 are roughly balanced (+23 vs.

Table 16: Decomposing the AOI gain into prompt-format vs. perception on DynaCU-Bench (100 tasks). **Standard** is a raw screenshot. **standard_structured** adds the AOI's structured observation-record prompt format (DOM element list, prior-step text trajectory) but no keyframes and no audio. **AOI full** is the full perception layer. Δ_{prompt} isolates the prompt-format effect; Δ_{percept} isolates the keyframe + audio + narration contribution. Gemini 3 Flash is reported separately in Table 20.

Model	**Standard**	**std_structured**	**AOI full**	$\boldsymbol{\Delta_{\text{prompt}}}$	$\boldsymbol{\Delta_{\text{percept}}}$
Claude Sonnet 4.6	38	**57**	82	+**19**	+**25**
GPT-5.4	37	**48**	57	+**11**	+**9**
Gemini 2.5 Flash	21	**46**	69	+**25**	+**23**
Open-source local models					
EvoCUA-32B	18	34	55	+16	+21
Fara-7B	17	**21**	34	+**4**	+**13**

Table 17: Sub-component decomposition of the prompt-format effect on Claude Sonnet 4.6 (100 dynamic tasks). Two new controls flank the existing `standard` and `standard_structured` configurations; "trajectory" covers prior-step action/text history and the structured sectioning.

Mode	**Pass/100**	**Trajectory?**	**`[PAGE EL]`?**
standard_minimal	16	no	no
standard	38	yes	no
standard_pageel_only	46	no	yes
standard_structured	**57**	yes	yes

+25 and +9 vs. +11 pp), and Fara-7B is sharply perception-leaning (+13 vs. +4 pp), consistent with the smallest model being reasoning-capacity-bottlenecked and unable to leverage a richer prompt format. Across all five models the worst-case framing that the entire AOI gain is a prompt-format artifact is decisively ruled out; the most favourable (perception is the only mechanism) is also too strong. For Gemini 2.5—which scored only 21/100 with raw screenshots—the prompt-format component alone recovers +25 pp: for video-native models, part of what the AOI buys is not new visual information but a better way to present the existing visual + textual context.

7.3 Which Prompt Sub-Component Carries the Load?

The `standard_structured` mode adds two pieces relative to the `standard` baseline: the `[PAGE ELEMENTS]` DOM-id list and the structured trajectory wrapping (`[CONTEXT]`/`[NEW]`/`[TASK]` sectioning around the prior-step text). To isolate which sub-component carries the load we run two further Claude Sonnet 4.6 controls: `standard_minimal` (task instruction plus screenshot only—no action history, no DOM list, no sectioning) and `standard_pageel_only` (the minimal prompt plus the `[PAGE ELEMENTS]` list). Table 17 reports the result.

Both sub-components carry load, sub-additively. The DOM-id list is the single largest lever: added to the minimal prompt it buys +30 pp ($16 \to 46$, McNemar $p = 1.5 \times 10^{-8}$), more than the trajectory alone ($16 \to 38$, +22 pp, $p = 4.8 \times 10^{-7}$). Neither alone reaches the combined effect: the trajectory still adds a significant +11 pp on top of the DOM list ($46 \to 57$, $p = 0.013$). The sub-additivity (+30 and +22 individually vs. +41 jointly) suggests the two partially overlap in function—both help the model track what it has already done and what on the page remains

Table 18: Additional runs on later-released models (100 dynamic tasks each). The Gemini 3 Flash run checks whether the Gemini 2.5 +48 pp gain replicates on the next-generation Gemini. Grok-4.3 is the latest Grok release; Grok-4-fast-reasoning is a lower-latency variant in the same family, included to control for the inference-latency floor identified in Section 8.1.

Model	**Standard**	**AOI full**	**Δ (pp)**	***p* (McNemar)**
Gemini 3 Flash	36	45	+9	0.18
Grok-4.3	25	65	+40	8.2e-10
Grok-4-fast-reasoning	19	47	+28	4.9e-06

Table 19: Grok-4 on DynaCU-Bench (100 dynamic tasks). The AOI delivers a significant +21 pp gain ($p = 3.0 \times 10^{-6}$), within the +17–+48 pp band reported in Table 3 for the other five models. Grok-4's absolute scores sit near the bottom of the range; manual log inspection shows this is driven by per-inference latency (∼80–180 s/call), which causes the agent to time out before completing more than a few steps on Medium and Hard tasks.

Mode	**Pass / 100**	**95% Wilson CI**	**Δ vs. Standard**	***p* (McNemar)**
Standard	4	[1.6, 9.8]	—	—
AOI full	25	[17.5, 34.4]	+21	3.0e-06

actionable.

8 Model Case Studies: Later-Released Models

Do the AOI's gains hold up on models released after the main-table runs? We ran three additional 100-task evaluations with the same harness: (i) **Gemini 3 Flash** [Google DeepMind, 2026], the next-generation successor to the Gemini 2.5 Flash result; (ii) **Grok-4.3**, the latest `grok-4` series model on xAI's API; (iii) **Grok-4-fast-reasoning**, a lower-latency variant included to isolate Grok-4's perception bottleneck from its inference-latency floor. Table 18 summarises the results. Together with the original Grok-4 run, these form two case studies: a model family whose binding constraint is inference *latency* rather than perception (Grok, Section 8.1), and a model on which one AOI component flips sign (Gemini 3, Sections 8.2–8.3).

8.1 The Grok Family: Latency as the Binding Constraint

The xAI Grok-4 series [xAI, 2026] was released after the initial Table 3 runs. We wire it through xAI's OpenAI-compatible vision endpoint (`api.x.ai/v1`) and run a full standard + AOI full eval on the 100 dynamic tasks (Table 19).

Grok-4's standard score (4/100) is the lowest of any model evaluated. Manual inspection shows it is not an action-grammar problem (the agent emits valid `wait`/`click`/`fill` actions) but a latency one: a single Grok-4 call routinely takes 80–180 s, so within the per-task wall-clock budget (§4) the agent completes only one or two steps. The +21 pp AOI gain therefore lower-bounds the true effect: the AOI succeeds on tasks where a few well-placed observations suffice (e.g. one call that captures the spoken answer), but cannot unblock latency-dominated tasks that require many sequential actions. Grok-4 is a case where inference latency, not perception, becomes the binding constraint once perception is unblocked.

Table 20: Decomposing the Gemini 3 Flash +9 pp AOI residual into prompt-format and perception components, using the same `standard_structured` protocol as Table 16. The Gemini 2.5 row is reproduced from Section 7.

Model	**Standard**	**std_structured**	**AOI full**	Δ_{prompt}	Δ_{percept}
Gemini 2.5 Flash	21	**46**	69	**+25**	**+23**
Gemini 3 Flash	36	**29**	45	**−7**	**+16**

Table 21: Gemini 3 Flash per-category success (pass/10), standard vs. AOI full, on the 100 dynamic tasks—the per-category breakdown underlying the Section 8.2 measurements (Gemini 3 is omitted from Table 6 for width). The four audio categories (A, B, G, H) gain +16 pp in aggregate; the shortfall vs. the Gemini 2.5 +48 pp gain is concentrated in the E and F regressions.

Mode	A	B	C	D	E	F	G	H	I	J	**Total**
Standard	0	2	4	4	**8**	3	1	7	5	2	36
AOI full	**6**	**6**	**6**	**5**	1	0	**6**	**8**	5	2	45
Δ	+6	+4	+2	+1	−7	−3	+5	+1	0	0	+9

Grok-4.3 vs. Grok-4-fast-reasoning: latency is real but not the whole story. The original Grok-4 sat at standard 4, AOI full 25 (attributed above to its ∼80–180 s/call inference latency). The two follow-on runs decompose that claim. Grok-4-fast-reasoning (same base model, lower latency) reaches AOI full 47, +22 pp over the original; Grok-4.3 reaches AOI full 65, a further +18 pp. So the gap to a "saturated" Grok ceiling decomposes into +22 pp from removing latency and +18 pp from base-model improvement: the latency-bound framing above was directionally right and is now quantitatively pinned.

8.2 Gemini 3 Flash: Scaffold-Policy Mismatch and a Keyframe Sign Flip

Gemini 3 Flash raises the standard score from 21 (Gemini 2.5) to 36 (+15 pp) and shrinks the AOI Δ from +48 to +9 pp ($p = 0.18$; Table 18). We initially read this small residual as the model having *internalised* the AOI's structured-record presentation. Three orthogonal measurements rule that out: the net residual is a three-way sum of an audio gain every other model also enjoys (+12), a scaffold effect that is bi-directional across categories (+9), and a Gemini-3-unique keyframe regression (−12).

Measurement 1: per-category AOI Δ. The aggregate +9 pp hides a stark within-model split. On the four audio categories where every CU model is fundamentally deaf (A podcast, B meeting, G phone, H interview), Gemini 3 gains $+6/+4/+5/+1 = +16$ pp—in line with Gemini 2.5 (+24 pp), Claude (+27 pp), and GPT-5.4 (+8 pp) on the same categories (Gemini 3 per-category numbers in Table 21; the other models in Table 6). On the four visual-temporal categories (C video, D carousel, I collab, J games) it gains $+2/+1/+0/+0 = +3$ pp, comparable to other models' +2 to +12. The entire shortfall vs. the +48 Gemini 2.5 gain sits on two categories, **E live dashboards** and **F transient UI**, where Gemini 3 alone *regresses* (−7 and −3 pp; no other model regresses there: Claude $+0/+5$, GPT-5.4 $+3/+6$, Gemini 2.5 $+8/+9$). Whatever shrinks the net residual lives entirely in this E+F regression, not in any global internalisation.

Measurement 2: action-grammar shift is universal; premature termination is Gemini-3-specific. The system prompt offers two ways to interact with a form input: visual `type("text")`, which auto-targets the focused element, and DOM-selector `fill(#id, "text")`, used when the

[PAGE ELEMENTS] block provides the id. Adding the [PAGE ELEMENTS] block shifts every cloud model toward fill() at similar rates: Claude 3→67/100 tasks, GPT-5.4 0→48, Gemini 2.5 3→66, Gemini 3 4→61. The grammar shift itself is therefore not the regression mechanism. What *is* unique to Gemini 3 is what happens *after* the fill: the rate of trajectories ending with a bare done action falls for every other model when given the scaffold (Claude 9→4, GPT-5.4 16→9, Gemini 2.5 14→9) but *rises* for Gemini 3 (8→12). Inspection of the failing traces (result_val of not_submitted; 8/100 in AOI full vs. 4/100 in standard) shows the same pattern: Gemini 3 fills the input via the listed selector, then emits done without explicitly clicking the submit button or pressing Enter.

Measurement 3: a four-way decomposition isolates each AOI component's per-model contribution. To distinguish "audio is the only missing content" from "Gemini 3 is internally saturated," and to separate the [PAGE ELEMENTS] scaffold effect from the keyframe-image effect, we run two additional controls on the full 100-task suite: (a) standard_audio: audio transcripts injected, but the [PAGE ELEMENTS] block suppressed (so action grammar and termination policy stay in the bare-standard regime); (b) aoi_audio: audio plus the full structured scaffold, but *no* keyframe images. With standard (36), standard_structured (29), and aoi_full (45), these form a four-way decomposition (Table 22) yielding three sign-separated component effects:

- *Audio:* $+12$ *pp* (standard 36 → standard_audio 48). Of this, $+10$ pp lives on the four audio categories alone, comparable in magnitude to the $+16$ pp aoi_full gets on the same categories. Gemini 3 is not audio-saturated.
- *Scaffold ([PAGE ELEMENTS] + structured trajectory wrapping):* $+9$ *pp* (standard_audio 48 → aoi_audio 57). Per-category the scaffold is *bi-directional* on Gemini 3: it adds $+15$ pp on the audio-Q&A categories where the listed input id resolves the targeting question (A: 3 → 9, B: 5 → 9, G: 4 → 8, H: 8 → 9), but it *subtracts* 4 pp on the E live-dashboard plus F transient-UI tasks (E: 6 → 1, F: 1 → 2) for the fill→done-without-submit reason traced in Measurement 2. The scaffold effect is also path-dependent: measured with audio present (this $+9$ pp) it differs from the audio-free path in Table 20 ($\Delta_{\text{prompt}} = -7$ pp, standard 36 → standard_structured 29), an audio×scaffold interaction rather than a contradiction; we report the audio-present value because aoi_full always includes audio.
- *Keyframes:* -12 *pp* (aoi_audio 57 → aoi_full 45). Not predicted by the prior-generation framing, which treated extra keyframes as monotonically additive perception. Removing keyframes helps on six of ten categories ($+3$ pp each on A, B, C; $+2$ on F, G; $+1$ on H), is neutral on E, I, and J (E already near floor from the scaffold cost), and costs -2 pp only on D—the carousel category, where keyframes do carry novel visual information—summing to the $+12$ pp total. Section 8.3 probes the mechanism causally.

The net $(+12) + (+9) + (-12) = +9$ pp matches the aoi_full residual, but reframes what the small net means: Gemini 3 has substantial slack the AOI can recover ($\sim$+21 pp with the right component mix, reaching 57/100 from 36); the default bundle is just mis-tuned for this model. One component (the keyframe images) crosses from net-positive on earlier models—$+10$ pp on Claude and $+6$ pp on Gemini 2.5, both realised through narration (Section 6.3)—to net-negative (-12 pp) on Gemini 3, while another (the structured scaffold) flips from uniformly positive to task-dependent.

8.3 Probing the Keyframe Regression Mechanism (Gemini 3 Flash)

The -12 pp keyframe effect on Gemini 3 (Section 8.2) is the first AOI component we observe flipping sign across model generations. Three candidate mechanisms could produce it: (a) *image-token dilution*, where extra image tokens dilute the model's attention regardless of what they show; (b) *novel-content distraction*, where the keyframes carry new visual information that conflicts with the model's preferred grounding signal (the post-action screenshot); (c) *a position effect*, where

Table 22: Gemini 3 Flash: four-way decomposition of the AOI residual. `aoi_audio` = audio enabled + page-elements scaffold + trajectory wrapping (no keyframes). `standard_audio` = audio enabled + standard prompt (no `[PAGE ELEMENTS]`, no trajectory wrapping). If the small AOI residual were a sign of internal saturation, `standard_audio` should be no better than `aoi_full`. Per-category numbers shown for the two categories where Gemini 3 regresses (E, F) and the four where audio is the irreplaceable signal (A, B, G, H).

Mode	**Total**	**A+B+G+H**	**E+F**	**[PAGE_EL]?**	**Audio?**	**Keyframes?**
standard	36	10	11	no	no	no
standard_structured	29	9	2	yes	no	no
standard_audio	48	20	7	**no**	**yes**	no
aoi_audio	**57**	35	3	yes	yes	no
aoi_full	45	26	1	yes	yes	yes

Table 23: Keyframe causal probe on Gemini 3 Flash: 50 visual-active tasks (categories B–F). Each mode keeps audio and the structured scaffold fixed and varies only the keyframe channel. The two reference anchors are the Section 8.2 runs restricted to the same subset. Every keyframe-bearing condition scores at or below the no-keyframe anchor, and the conditions are pairwise indistinguishable (McNemar $p \geq 0.375$).

Mode	**Pass / 50**	**Rate**	**Probes**
aoi_audio (no keyframes, anchor)	**24**	48%	—
aoi_full_max1kf	20	40%	budget 1
aoi_full_max2kf	18	36%	budget 2
aoi_full_max3kf	20	40%	budget 3
aoi_full (= max5kf, anchor)	18	36%	default budget 5
aoi_full_noise_kf	17	34%	image tokens, zero content
aoi_full_dup_kf	20	40%	image tokens, duplicate content
aoi_full_reorder_kf	22	44%	keyframes after screenshot

keyframes placed before the screenshot displace the recency signal the model uses for action targeting. We disentangle these on the 50-task subset where keyframes are actively produced (categories B, C, D, E, F; A, G, and H produce near-zero keyframes, and I and J are dominated by interaction effects). Every probe mode keeps the audio and structured-scaffold pipeline of `aoi_full` fixed and varies only the keyframe channel: `max{1,2,3}kf` cap the per-step keyframe budget below the default 5; `noise_kf` keeps the keyframe count but replaces each keyframe with a random-noise image of identical size; `dup_kf` replaces each keyframe with a copy of the post-action screenshot; `reorder_kf` moves the keyframes after the post-action screenshot.

Table 23 gives a clear answer: *the regression is caused by the presence of extra images per se, not by their content, count, or position.* Three observations pin the mechanism. (i) A single keyframe per step (`max1kf`, 20/50) already pays most of the penalty, and the budget sweep from 1 to 5 keyframes is non-monotonic (20, 18, 20, 18)—the cost does not scale with image count. (ii) `noise_kf` (17/50), whose keyframes carry *zero* task information, is indistinguishable from the real-keyframe conditions (17–20/50; e.g. `max3kf` vs. `noise_kf`: $p = 0.45$)—ruling out novel-content distraction (b): the model is not confused by what the keyframes show, it is penalised for their presence. (iii) `dup_kf` (20/50) and `reorder_kf` (22/50) recover little, so position (c) is at most a minor modifier. The picture is image-token dilution (a) with a roughly fixed cost for *any* extra image: Gemini 3's action grounding degrades when the prompt contains images beyond the post-action screenshot, regardless of what they show. At N=50 the pairwise differences among probe conditions are underpowered, so we treat the mechanism identification as strong directional evidence rather than a definitive ranking; the actionable consequence, however, is unambiguous and already followed from Section 8.2: on Gemini 3 the keyframe channel should be disabled (`aoi_audio`), and no cheaper keyframe variant rescues it.

8.4 Synthesis: A Diagnostic Map of Model Slack

The observation interface is not a monolith, and its components flip sign between model generations. Pulling the case studies together with the Section 7 decompositions (Δ_{prompt}, Δ_{percept}) yields a per-model map:

- *Gemini 3 Flash:* not perception- or audio-saturated. Net +9 is a three-way cancellation (+12 audio, +9 scaffold with bi-directional per-category effects, −12 keyframes). Audio+scaffold-only (`aoi_audio`) reaches 57/100—a +21 pp recovery, +12 above `aoi_full`. Components must be tuned per-model rather than bundled.
- *Gemini 2.5 Flash:* +25, +23. Both prompt-format and perception useful; neither internalised. Within perception the keyframe images add +6 pp, and only through narration (Section 6.3).
- *Claude Sonnet 4.6:* +19, +25. Both useful; perception the larger lift. Within perception the keyframe images contribute +10 pp—but only via narration (+1 pp as raw input); audio and narration carry the rest (Section 6.3).
- *Fara-7B:* +4, +13. Reasoning-capacity-bottlenecked; the model cannot leverage richer prompt presentation.
- *Grok-4 (original):* latency-bottlenecked. The AOI full ceiling of 25 decomposes (via Grok-4-fast vs. Grok-4.3) into ∼+22 from removing latency and ∼+18 from base-model improvement.

The AOI's residual gain still functions as a diagnostic map of where a model has slack, but the dimensions are richer than "perception only": perceptual, prompt-handling, latency, reasoning-capacity, *and* action-policy compatibility with the scaffold. Calibrating the scaffold to the target model's native action grammar is the natural next step.

9 Comparison with Streaming Multimodal Baselines

A natural alternative to the AOI is a *streaming, end-to-end multimodal model* that bundles perception with reasoning into a single trained model. The two production endpoints exposing this paradigm today are Gemini Live [Google, 2025] (audio + video streams + tool calls) and the OpenAI Realtime API [OpenAI, 2024] (audio + tool calls; vision via the standard chat endpoint). Neither is shipped as a CU agent: each is a voice-first assistant with function-calling. We adapt them to DynaCU-Bench by opening one session per task, feeding in the page audio captured via PulseAudio, sending screenshots as multimodal turns, and translating returned tool calls into browser actions (implementation: `aoi/realtime_baselines.py`). The Gemini baseline uses `gemini-2.5-flash-native-audio`, the highest function-calling-capable Live tier, with audio streamed natively into the session. For OpenAI, no Realtime tier we probed at evaluation time (May 2026; `gpt-realtime` beta and `gpt-realtime-2.0` GA) accepts vision tokens *inside* the websocket session, so our primary baseline follows the integration path the OpenAI Realtime cookbook recommends for "giving a voice assistant eyes": chat-completions on `gpt-4o` (the chat-endpoint twin of `gpt-4o-realtime-preview`), with captured page audio accumulated as transcript text, the current screenshot as an image turn, and forced tool choice. Routing the same adapter through the newer `gpt-realtime-2.0` model id scores 0/12 on the audio subset (and 0/5 on the sanity set below), so the weak result is not specific to the gpt-4o backbone.

Table 24: Comparison against monolithic streaming baselines on a 12-task balanced cross-category subset (3 tasks each from podcast A, meeting B, phone G, interview H). We adapt each streaming API to DynaCU-Bench by opening one session per task, streaming page audio + screenshots in, and translating returned tool calls into browser actions. Each baseline is scored on the same 12 task IDs the AOI is scored on in Table 3. OpenAI backbone and Gemini Live tier choices are detailed in the body below.

Configuration	A	B	G	H	Total / 12
OpenAI Realtime adapter (gpt-4o backbone)‡	0	0	0	3	3 (25%)
Gemini Live (2.5 flash native-audio)	0	0	0	0	0 (0%)
AOI full (Claude Sonnet 4.6), same tasks	3	3	3	3	**12** (100%)

‡ The Realtime websocket does not accept vision tokens in the same session, so vision and transcribed audio are supplied through chat-completions on gpt-4o (backbone note in the body). The gpt-realtime-2.0 backbone scores 0/12 on the same subset.

Adapter sanity check. To rule out broken-adapter as an explanation for the audio-subset numbers, we evaluate both baselines on a five-task purely-visual sanity set across four categories (C-E1, E-E1, F-E1, F-E2, I-E1; two F tasks cover both transient-banner subtypes). Each adapter emits valid tool calls of every supported type (`click`, `fill`, `type_text`, `scroll`); the browser executes them; the resulting DOM is scored by the main benchmark's success criterion (Table 25; per-task detail in Appendix F). Both baselines hit action-grammar success on 5/5 tasks but task-correct success on 0/5: clicks land on the wrong UI elements, typed text is hallucinated, and the model commits before reading the page. On the cookie-banner task (F-E2), OpenAI Realtime clicks "dismiss" instead of "accept"—the click reached the DOM (`cookies_dismissed` is recorded), but the model chose the wrong target. AOI-equipped Claude Sonnet 4.6 passes 4/5 on the same sanity set, confirming the tasks are solvable through a vision-grounded CU loop. The audio-subset failures below therefore reflect intrinsic limitations of streaming-voice-with-tool-calls on grounded computer-use, not adapter scaffolding.

Both streaming baselines underperform the AOI by a wide margin on this audio-focused subset. OpenAI Realtime succeeds only on the simplest interview tasks (3/3), where the user's question is short enough to answer from a single audio chunk; on podcast and meeting tasks it emits `wait()` tool calls indefinitely without ever extracting and acting on the spoken content. Gemini Live scores zero: manual inspection shows it returns very few tool calls per session, and when it does, it tends to issue clicks at non-actionable coordinates rather than commit to a typed answer—a model optimised for verbal-response use cases (audio in, audio out) rather than grounded function-call decisions on a screenshot. Together with the adapter sanity check above, the picture validates the central design hypothesis: *a streaming voice model with function-calling is not a substitute for a perception layer that persists what it heard as text in the trajectory.*

10 Related Work

Computer-use agents. The CU agent landscape spans closed-source systems [Anthropic, 2024a, OpenAI, 2025, Google DeepMind, 2025, Andreux et al., 2025] and open-source models [Wang et al., 2025a,b, Awadallah et al., 2025, Ye et al., 2025, Song et al., 2025]. Agent S3 [Simular AI, 2025] demonstrates competitive open-source performance on OSWorld. A recent CU agent survey [Sager et al., 2025] catalogues many agents and datasets and identifies several open gaps, but does not identify the observation modality as one. All of these agents operate through the screenshot–reason–act loop; none addresses dynamic visual content or audio perception.

Table 25: Adapter sanity check for the streaming baselines. Five purely-visual tasks (no audio comprehension required). **Adapter exercised**: counts the tasks where the model emitted at least one valid tool call and the browser executed it (action-grammar success). **Task-correct**: counts the tasks where the resulting DOM passed the benchmark's success check. The adapter is fully exercised on every task; task-correct failures are content errors (wrong button, wrong text, hallucinated answer), not infrastructural.

Baseline	Adapter exercised	Task-correct (visual sanity)	Audio subset (12 tasks)
OpenAI Realtime adapter (gpt-4o backbone)	5/5	0/5	3/12
Gemini Live (2.5 flash native-audio)	5/5	0/5	0/12
AOI full (Claude Sonnet 4.6) reference	5/5	4/5	12/12

Real-time multimodal models. Gemini Live [Google, 2025] and the OpenAI Realtime API [OpenAI, 2024] process continuous audio/video/text streams in real time, bundling perception with reasoning into a single trained model. Our work is complementary: the AOI decouples perception from reasoning, so the same perception layer benefits any CU model without retraining. Section 9 compares the two paradigms head-to-head on the audio-focused DynaCU-Bench subset.

Video understanding for GUI. GUI-World [Chen et al., 2024], VideoWebArena [Jang et al., 2024], and CUA-Suite [Chen et al., 2025] identify the observation problem through benchmarking. MemGUI-Bench [Park et al., 2025] reveals memory gaps in GUI agents. These benchmarks identify the problem but do not propose solutions compatible with existing CU agents.

Keyframe selection. VideoTree [Wang et al., 2024] and AKS [Tang et al., 2025] use CLIP-based frame selection for offline video QA. Apollo [Zohar et al., 2024] finds that FPS sampling outperforms uniform sampling and identifies optimal token budgets. All prior work targets offline video understanding; our contribution is real-time observation filtering for CU agents, where the two-stage design (sub-millisecond pixel gate followed by $\sim$7 ms CLIP only on changed frames) keeps the per-frame cost low enough to run continuously in the background between agent steps.

Adaptive middleware for GUI agents. Cradle [Tan et al., 2024] includes frame difference analysis for games—the closest predecessor to the AOI's perception layer—but uses pixel-level differencing without semantic filtering and has no audio processing. ScreenLLM [Jin et al., 2025] captures stateful screen schemas with keyframe extraction, complementing visual narration but without audio or adaptive observation. StreamAgent [Yang et al., 2025] and Dispider [Qian et al., 2025] separate perception from response generation for streaming video, paralleling the AOI's architecture, but neither addresses CU agent observation.

Structured agent interfaces. A complementary line of work bypasses the perception problem by having websites or applications expose structured interfaces directly to agents: the Model Context Protocol [Anthropic, 2024b] for tool/data exchange and declarative web specifications such as NLWeb [Microsoft, 2025] for site-side agent endpoints. These approaches are powerful where adoption exists, but require website cooperation; the AOI enables perception on arbitrary, unmodified web content (any HTML/audio rendered to a browser, including legacy or third-party pages with no agent endpoint).

11 Limitations

Action latency is unsolved. The AOI extends perception but does not accelerate decision-making. CU models still require 1–5 seconds per inference step, making the approach unsuitable for real-time games or fast interactive applications (as evidenced by category J results).

Temporal resolution floor ($\sim$300 ms). Sampling at $\sim$3 Hz means events shorter than $\sim$300 ms can fall entirely between samples. This primarily affects category J (browser games) where targets appear briefly. For human-paced computer use (meetings, web browsing, presentations) this resolution is sufficient; for high-speed visual content, information is irretrievably lost.

Context-window pressure on keyframe-heavy tasks. On categories D and E, accumulated keyframe images can exceed the CU model's context window—we observed overflow at 8,192 tokens on some local models. Retaining text narrations rather than images in long trajectories (Section 6.2) sidesteps this almost entirely.

Reasoning capacity remains a separate bottleneck. For smaller models (Fara-7B), improved perception does not help with tasks requiring multi-step reasoning, arithmetic, or complex instruction following: better inputs do not improve reasoning over those inputs. Fara-7B's gain on hard tasks is small ($3 \to 4$) compared to easy tasks ($9 \to 19$).

Statistical power. Each cell in our main table is one trial of 100 binary tasks; we use paired McNemar's exact tests to assess differences but the data is not powered for fine-grained ranking between configurations whose marginal scores are within a few percentage points.

Persistent-memory effect smaller than the headline. The narration-persistence gap is significant at $N{=}100$ ($p = 0.039$), but the content audit of the ten discordant tasks in Section 6.2 finds prior narrations literally containing the load-bearing fact in only two of the ten wins; the rest operate through trajectory-level priming and, on first-step wins, decoding variance. Persistent memory acts as literal recall on a minority of genuinely multi-step tasks, not as a uniform lift, and a larger benchmark with more memory-heavy tasks would be needed to measure the memory effect in isolation.

Narration quality is bounded by the CU model. Long-term visual memory depends on the CU model producing accurate text descriptions. Errors—a misread number, a hallucinated UI element—are permanently encoded in the trajectory.

Audio limited to speech transcription. Our implementation uses Whisper for speech-only ASR rather than a multimodal audio model. Non-speech audio events (notification sounds, alert beeps) are not transcribed. A multimodal audio model (e.g., Qwen3 Omni) could extend coverage to the full audio scene.

Benchmark scope and external validity. DynaCU-Bench tasks execute in a controlled browser environment with synthesized audio. Performance on real-world dynamic content (live video calls, real YouTube, native desktop applications) may differ. We also do not report AOI numbers on existing dynamic-GUI benchmarks (GUI-World [Chen et al., 2024], VideoWebArena [Jang et al., 2024], CUA-Suite [Chen et al., 2025], VideoGameBench [Zhang et al., 2025]). These either evaluate offline video question-answering over pre-recorded clips rather than a live screenshot–action loop (GUI-World, CUA-Suite), or supply video as fixed context to a web agent rather than a continuously observed environment with a separate audio channel (VideoWebArena), so wrapping them with the AOI's inter-step capture and volume-gated audio would require re-instrumenting each harness around our observe–act loop. Because every headline number here comes from our own benchmark, this re-instrumentation is the most direct external-validity test of the AOI and is the clearest item of future work.

Open-source replication on a substitute model. For the selection-method convergence finding we additionally evaluate Qwen3-VL-32B-Instruct [Qwen Team, 2025] via OpenRouter as an independent open-source vision model. The convergence pattern reproduces (Section 6.1, Table 10), and the main standard-vs.-AOI contrast replicates on two further Qwen3-VL models (Table 4).

Real-content task design. The DynaCU-Real-Local harness (Appendix E) verifies cross-engine ASR robustness, but its tasks have answers recoverable from the CU model's pretraining, so they cannot also serve as a perception comparison. Designing real-recording tasks whose ground-truth answers are not in the model's pretraining is harder than it sounds and is left to future work.

12 Conclusion

Computer-use agents cannot perceive dynamic content because observation is tied to action: one screenshot per step. We argue the agent-computer *observation* interface is a separable, underexplored design axis—the natural counterpart to the action interface SWE-agent [Yang et al., 2024] surfaced—and introduce the AOI, a perception layer transparent for static tasks, adaptive for dynamic content, and compatible with any existing CU model without retraining. Eight CU agents equipped with the AOI—six in the main table plus a two-model Qwen3-VL replication—gain +17 to +48 pp on DynaCU-Bench (paired McNemar $p < 10^{-3}$ for every model); the strongest, Claude Sonnet 4.6, reaches a 3-seed mean of 78.7% (std 3.1 pp; best seed 82%).

Three findings reframe how we think about CU perception. First, keyframe *selection* does not significantly affect accuracy—uniform, random, pixel-difference, and CLIP-based methods are statistically indistinguishable (56–58% on Claude, replicated on the open-source Qwen3-VL-32B)—so no learned or semantic frame selector is needed; and the keyframe *images* pay off only through narration, adding +1 pp as raw visual input but +10 pp once the model narrates the captured frames into persistent text (Claude, Section 6.3). Second, among the AOI's content components the largest single contributor is visual narration (+18 pp): roughly +10 pp from inference-time chain-of-thought during narration generation plus +8 pp from persisting the narration as text, which a content audit attributes to literal memory recall on a minority of multi-step tasks and to weaker trajectory-level priming elsewhere. Narration's primary mechanism is making the model articulate what it sees; persistence matters where information genuinely accumulates. Third, the observation interface is not a fixed bundle: on Gemini 3 Flash the keyframe stream—never regressive on the earlier model generations we measured—flips to −12 pp, a regression that a causal probe attributes to image-token dilution rather than content (Section 8.3); audio and the structured scaffold stay positive, and dropping keyframes recovers +21 pp over standard. Observation-interface components must be selected per model.

Three further results sharpen the picture. Token cost falls with the AOI on every cloud model (15–50%) because better perception yields fewer steps per task; wall-clock time rises for fast-inference models and falls for slow ones. A direct `standard_structured` control on DynaCU-Bench separates the AOI's prompt-format component from its perception component for five models (Section 7; Gemini 3 Flash in Section 8.2), ruling out the worst-case confound that the gain is entirely a prompt-format artifact. On a 12-task audio-focused subset, production streaming baselines—OpenAI Realtime (3/12) and Gemini Live (0/12)—underperform AOI (12/12) by a wide margin: streaming voice models with function-calling are not a substitute for a perception layer that persists what was heard.

We release the entire stack—perception layer, DynaCU-Bench, Static-50, DynaCU-Real-Local, and streaming-baseline adapters—so future CU research can cleanly decouple observation gains from model-side gains.

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A Full Per-Category Results

Table 26: Complete per-category results for all 8 observation configurations tested with Claude Sonnet 4.6.

Mode	A	B	C	D	E	F	G	H	I	J	**Total**
Standard	0	2	4	4	8	4	1	7	3	5	38
Uniform 1 FPS	0	4	10	8	9	8	1	8	6	4	58
Uniform 3 FPS	2	5	8	8	8	8	1	7	6	3	56
Pixel-diff	1	5	9	7	9	8	1	8	6	4	58
Random keyframes	1	6	8	6	9	8	1	8	6	3	56
AOI visual	1	5	9	7	8	9	1	8	6	4	58
AOI visual+ASR	1	4	10	9	9	9	4	9	6	3	64
AOI full	**9**	**10**	**9**	**9**	**8**	**9**	**9**	**9**	**8**	2	**82**

B DynaCU-Bench Task List

Each of the 100 tasks is identified by a category letter (A–J) and a difficulty-indexed ID. Easy tasks: E1, E2, E3. Medium tasks: M1, M2, M3, M4. Hard tasks: H1, H2, H3.

Example tasks by category:

- **A-E1** (Podcast, Easy): Listen to a podcast segment and identify the guest's name.
- **B-M2** (Meeting, Medium): During a meeting with slides, count the number of items in a presented chart.
- **C-H1** (Video, Hard): Watch a terminal recording and describe the full workflow demonstrated.
- **D-M4** (Carousel, Medium): Identify a specific product name from a rotating carousel.
- **E-E3** (Dashboard, Easy): Read the current warning level from a live-updating dashboard.
- **F-E2** (Transient UI, Easy): Click "Accept" on a cookie consent banner before it auto-dismisses.
- **G-M1** (Phone, Medium): Follow spoken instructions during a simulated phone call.
- **H-H2** (Interview, Hard): Complete a multi-question spoken interview with scoring.
- **I-M2** (Collab, Medium): Resolve a conflict in a collaborative document when a remote edit arrives.
- **J-E2** (Games, Easy): Complete 3 rounds of a simple reaction-time game.

All tasks are implemented as self-contained HTML files with embedded JavaScript for task logic, audio synthesis, and evaluation. The complete benchmark (100 HTML files, evaluation scripts, and task definitions) will be released publicly.

C CLIP Threshold Calibration

The default threshold $\theta = 0.04$ was calibrated empirically. Web UI events such as modal dialogs appearing produce cosine distances of $\sim$0.08; video scene cuts produce distances of $\sim$0.15–0.25; loading spinners and cursor blinks produce distances of $\sim$0.005–0.02. Setting $\theta = 0.04$ captures all meaningful UI changes while filtering most periodic noise.

Figure 9 shows the CLIP threshold θ sweep across a 15$\times$ range from 0.02 to 0.30, evaluated on the 40 visual tasks (categories C–F) where keyframe selection is relevant. Point estimates remain within an 80–90% band throughout, demonstrating that for practitioners who choose the CLIP-based variant, threshold selection is not critical. There is no cliff: even at $\theta = 0.30$ (very aggressive

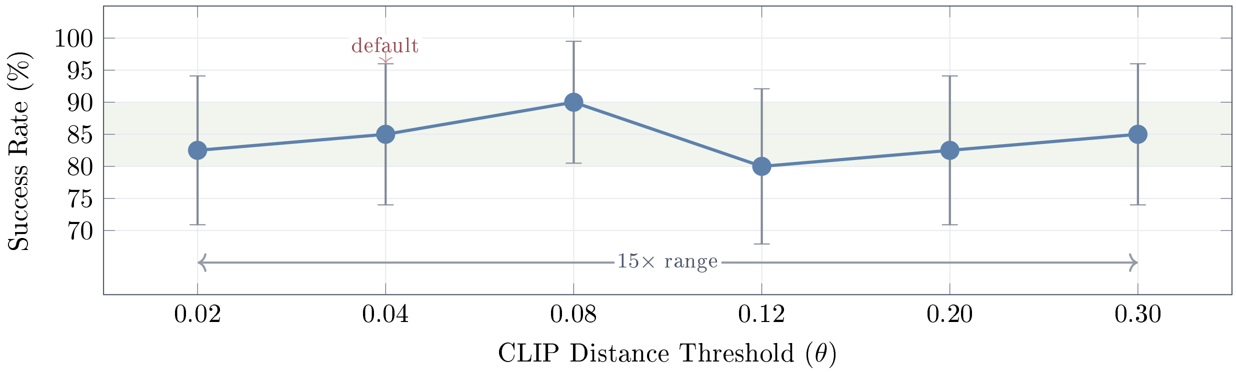

Figure 9: **CLIP threshold (θ) sensitivity** on 40 visual tasks (categories C–F) with Claude Sonnet 4.6. Error bars are 95% Wilson confidence intervals at N=40 (±9.5–12.1 pp). Point estimates remain within an 80–90% band (shaded) across a 15× range of θ, but the CIs overlap heavily, so we cannot distinguish individual thresholds at this sample size. θ values are placed at uniform spacing on the axis (not to scale). The bimodal distribution of CLIP distances—near-zero for unchanged content, large for meaningful changes—makes the method naturally insensitive to θ.

filtering that captures only major visual changes), the point estimate remains at 85%. At $\theta = 0.02$ (very sensitive, capturing small changes), it is 82.5%. The point estimate peaks at $\theta = 0.08$ (90%), but with N=40 the 95% Wilson confidence intervals on every point span ∼10–12 pp and overlap heavily—we therefore do not claim a true sweet spot, only that the method is robust across a 15× range of θ and that any reasonable choice will work. This insensitivity arises because CLIP embeddings produce a bimodal distribution of distances: semantically unchanged frames cluster near zero, while meaningful changes produce large distances, with relatively few samples in between.

D Implementation Details

Software. Python 3.11, Playwright 1.49 for browser automation, PulseAudio 16.1 for audio I/O, edge-TTS for high-quality speech synthesis, OpenAI CLIP ViT-B/16 for keyframe extraction, Whisper large-v3 for speech transcription, vLLM 0.19.0 for local model serving.

Hardware. All experiments run on a single machine with an NVIDIA RTX PRO 6000 Blackwell GPU (96 GB VRAM), 192 GB RAM. Cloud models (Claude, GPT-5.4, Gemini 2.5 Flash) are accessed via HTTP APIs. Local models (EvoCUA-32B, Fara-7B) are served via vLLM with 4-bit quantization (`bitsandbytes`). Whisper runs on CPU to avoid GPU contention. Running the two local models in the `standard_structured` mode (Section 7.2) required patching vLLM 0.19.0 around an NVML/userspace driver mismatch on our host: the CUDA platform plugin trusts `pynvml.nvmlInit()` alone, and an overly strict free-memory assertion fired when the Whisper service released GPU memory during startup profiling. Both patches are released with the code.

Hyperparameters. Screen capture rate: ∼3 Hz. Pixel change threshold (α): 1%. CLIP distance threshold (θ): 0.04. Maximum keyframes per step: 5. Audio capture: a 70 s ring buffer at 16 kHz mono holds rolling history; when the volume gate fires, Whisper is invoked once per step on the new inter-step audio (typically ∼3.5 s) extended by a ∼3.5 s overlap into the previous interval (∼7 s total) for boundary continuity. Post-action wait: 2.0 s. Maximum steps per task: 15.

Table 27: DynaCU-Real-Local: Claude Sonnet 4.6 Standard vs. AOI full on 12 real-recording tasks. Both modes tie at 11/12 on this set; the failure is the same task (`R_cast2_pip`) for both.

Mode	**Podcast**	**Meeting**	**Screencast**	**Voice**	**Total / 12**
Standard	3/3	3/3	2/3	3/3	**11** (92%)
AOI full	3/3	3/3	2/3	3/3	**11** (92%)

Table 28: Per-task adapter sanity-check outcomes for each streaming baseline. Both adapters drive the browser and emit valid tool calls (`click`, `fill`, `type_text`) on every task; every attempted action lands on the DOM. Both score 0/5 task-correct because the models choose the wrong target (e.g., F-E2 cookies dismissed instead of accepted), hallucinate content (C-E1 typed a URL not present in the recording), or commit prematurely (I-E1 typed before reading the page). The same failure modes appear on the audio subset (Table 24), confirming the audio-subset failures are content-driven rather than infrastructural.

Baseline	**C-E1**	**E-E1**	**F-E1**	**F-E2**	**I-E1**	**Total**
OpenAI Realtime adapter (gpt-4o backbone)	✗	✗	✗	✗	✗	0
Gemini Live (2.5 flash native-audio)	✗	✗	✗	✗	✗	0
AOI full (Claude Sonnet 4.6) reference	✓	✗	✓	✓	✓	4

E DynaCU-Real-Local Details

The 12 tasks span four sub-domains: 3 podcast (Aesop animal-identification), 3 meeting (Python / Postgres / Rust technical detail), 3 screencast (`git clone` / `pip install` / `npm test` outcome), and 3 voice (yes/no / directions / appointment day). Audio for podcast/meeting/voice tasks is rendered with `espeak` (a formant synthesizer with a deliberately non-neural acoustic profile). Screencast tasks use real `asciinema` v2 cast files generated locally. Source texts are public-domain materials: Aesop's Fables for podcasts, Wikipedia for Python language facts, project documentation for PostgreSQL/MVCC and Rust/tokio. All HTML harnesses, asset-build scripts, and success checks are released with the benchmark.

F Streaming-Baseline Adapter Sanity Check Details

To rule out the explanation that the streaming baselines' weak audio-subset numbers (Section 9) are an artifact of our adapter rather than a model limitation, we evaluate both adapters on a small purely-visual sanity set of five tasks across four categories: C (video / static recording), E (live dashboard), F (transient UI; two tasks, one banner-accept and one cookie-accept subtype), and I (collaborative document). All chosen so the answer can be reached without any audio comprehension. The exact set is C-E1, E-E1, F-E1, F-E2, and I-E1. For each baseline, we open one streaming session per task, send the post-action screenshot as a multimodal turn, and translate any returned tool calls into browser actions through the same path used in the main audio-subset evaluation. The success criterion is the same DOM check as the main benchmark. Table 28 reports the per-task outcome and the modal failure mode where applicable.